# CDx4HC405x, CD4HCT405x High-Speed CMOS Logic Analog Multiplexer and Demultiplexer

## 1 Features

- Wide analog input voltage range: ±5V maximum
- Low ON-resistance:
  - 70Ω typical ($V_{CC}$ – $V_{EE}$ = 4.5V)
  - 40Ω typical ($V_{CC}$ – $V_{EE}$ = 9V)
- Low crosstalk between switches
- Fast switching and propagation speeds
- Break-before-make switching
- Wide operating temperature range: –40°C to +125°C
- Operation control voltage: 4.5V to 5.5V
- Switch voltage: 0V to 10V
- Direct LSTTL input logic compatibility $V_{IL}$ = 0.8V maximum, $V_{IH}$ = 2V minimum
- CMOS input compatibility $I_I \leq 1\mu A$ at $V_{OL}$, $V_{OH}$

## 2 Applications

- Digital radio
- Signal gating
- Factory automation
- Televisions
- Appliances
- Programmable logic circuits
- Sensors

## 3 Description

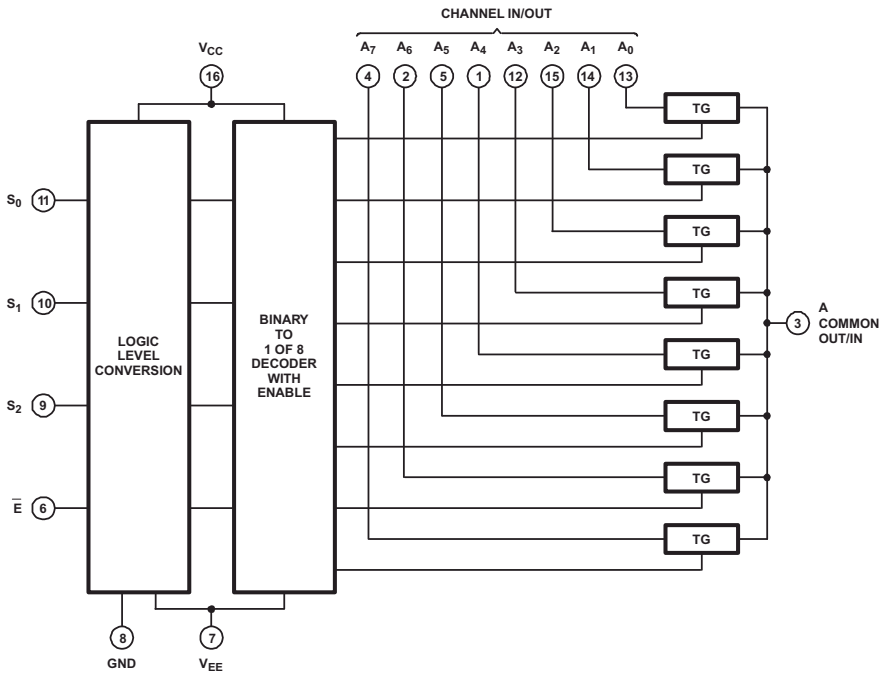
The CDx4HC405x and CDx4HCT405x device is a digitally controlled analog switch that uses silicon gate CMOS technology to achieve operating speeds similar to LSTTL with the low-power consumption of standard CMOS integrated circuits.

This analog multiplexer and demultiplexer controls analog voltages that may vary across the voltage supply range (for example, $V_{CC}$ to $V_{EE}$). It is a bidirectional switch that allows any analog input to be used as an output and vice versa. The switch has low ON resistance and low OFF leakages. In addition, this device has an enable control that, when high, disables all switches to their OFF state.

**Device Information**

| PART NUMBER | $T_A$ | PACKAGE(1) | PACKAGE SIZE(2) |
|---|---|---|---|
| CD54HCx405x | -55°C to 125°C | J (CDIP, 16) | 19.56mm × 6.92mm |
| CD74HCx405x | | N (PDIP, 16) | 19.30mm × 6.35mm |
| | | D (SOIC, 16) | 9.9mm × 3.9mm |
| | | NS (SOP, 16) | 10.3mm × 5.3mm |
| | | PW (TSSOP, 16) | 5mm × 4.4mm |

(1) For more information, see Section 11.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.




**Functional Diagram of HCT4051**

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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# 4 Pin Configuration and Functions

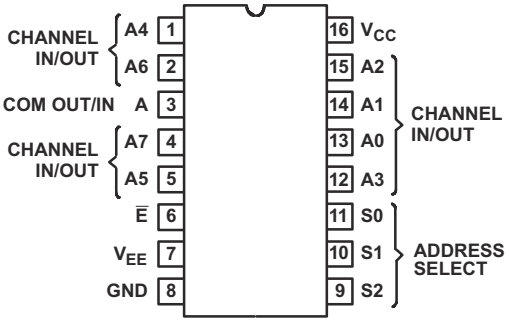



**Figure 4-1. CDx4HCx4051 J, N, D, NS, PW Packages 16-Pin CDIP, PDIP, SOIC, SO, TSSOP (Top View)**

**Table 4-1. Pin Functions for CDxHCx4051B**

| PIN | | TYPE(1) | DESCRIPTION |
|---|---|---|---|
| NAME | NO. | | |
| CH A4 IN/OUT | 1 | I/O | Channel 4 in/out |
| CH A6 IN/OUT | 2 | I/O | Channel 6 in/out |
| COM OUT/IN | 3 | I/O | Common out/in |
| CH A7 IN/OUT | 4 | I/O | Channel 7 in/out |
| CH A5 IN/OUT | 5 | I/O | Channel 5 in/out |
| !E | 6 | I | Enable Channels (Active Low) |
| $V_{EE}$ | 7 | — | Negative power input |
| GND | 8 | — | Ground |
| S2 | 9 | I | Channel select 2 |
| S1 | 10 | I | Channel select 1 |
| S0 | 11 | I | Channel select 0 |
| CH A3 IN/OUT | 12 | I/O | Channel 3 in/out |
| CH A0 IN/OUT | 13 | I/O | Channel 0 in/out |
| CH A1 IN/OUT | 14 | I/O | Channel 1 in/out |
| CH A2 IN/OUT | 15 | I/O | Channel 2 in/out |
| $V_{CC}$ | 16 | — | Positive power input |

(1) I = input, O = output

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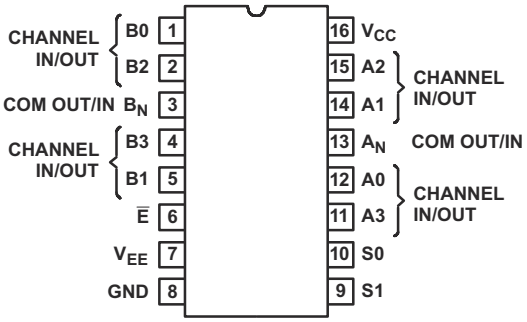



**Figure 4-2. CDx4HCx4052 J, N, D, NS, PW Packages 16-Pin CDIP, PDIP, SOIC, SO, TSSOP (Top View)**

**Table 4-2. Pin Functions for CDx4HCx4052B**

| PIN | | TYPE(1) | DESCRIPTION |
|---|---|---|---|
| NAME | NO. | | |
| CH B0 IN/OUT | 1 | I/O | Channel B0 in/out |
| CH B2 IN/OUT | 2 | I/O | Channel B2 in/out |
| COM B OUT/IN | 3 | I/O | B common out/in |
| CH B3 IN/OUT | 4 | I/O | Channel B3 in/out |
| CH B1 IN/OUT | 5 | I/O | Channel B1 in/out |
| !E | 6 | I | Enable channels (Active Low) |
| $V_{EE}$ | 7 | — | Negative power input |
| GND | 8 | — | Ground |
| S1 | 9 | I | Channel select 1 |
| S0 | 10 | I | Channel select 0 |
| CH A3 IN/OUT | 11 | I/O | Channel A3 in/out |
| CH A0 IN/OUT | 12 | I/O | Channel A0 in/out |
| COM A IN/OUT | 13 | I/O | A common out/in |
| CH A1 IN/OUT | 14 | I/O | Channel A1 in/out |
| CH A2 IN/OUT | 15 | I/O | Channel A2 in/out |
| $V_{CC}$ | 16 | — | Positive power input |

(1) I = input, O = output


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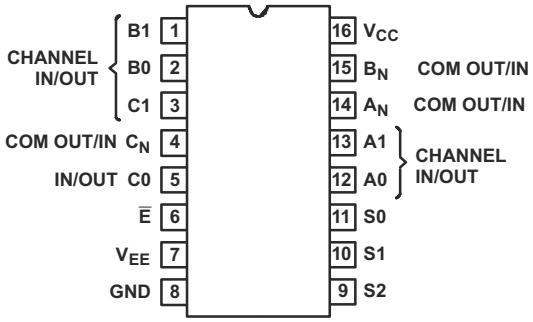



**Figure 4-3. CDx4HCx4053 J, N, D, NS, PW Packages 16-Pin CDIP, PDIP, SOIC, SO, TSSOP (Top View)**

**Table 4-3. Pin Functions CDx4HCx4053B**

| PIN | | TYPE[1] | DESCRIPTION |
|---|---|---|---|
| NAME | NO. | | |
| B1IN/OUT | 1 | I/O | B channel Y in/out |
| B0 IN/OUT | 2 | I/O | B channel X in/out |
| C1 IN/OUT | 3 | I/O | C channel Y in/out |
| COM C OUT/IN | 4 | I/O | C common out/in |
| C0 IN/OUT | 5 | I/O | C channel X in/out |
| !E | 6 | I | Enable channels (Active Low) |
| $V_{EE}$ | 7 | — | Negative power input |
| GND | 8 | — | Ground |
| S2 | 9 | I | Channel select 2 |
| S1 | 10 | I | Channel select 1 |
| S0 | 11 | I | Channel select 0 |
| A0 IN/OUT | 12 | I/O | A channel X in/out |
| A1 IN/OUT | 13 | I/O | A channel Y in/out |
| COM A OUT/IN | 14 | I/O | A common out/in |
| COM B OUT/IN | 15 | I/O | B common out/in |
| $V_{CC}$ | 16 | — | Positive power input |

(1) I = input, O = output

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# 5 Specifications

## 5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

| | | | MIN | MAX | UNIT |
|---|---|---|---|---|---|
| $V_{CC} - V_{EE}$ | DC Supply voltage | | -0.5 | 10.5 | V |
| $V_{CC}$ | | | –0.5 | 7 | V |
| $V_{EE}$ | | | 0.5 | -7 | V |
| $I_{IK}$ | DC input diode current | $V_I < – 0.5V$ or $V_I > V_{CC} + 0.5V$ | –20 | 20 | mA |
| $I_{OK}$ | DC switch diode current | $V_I < V_{EE} – 0.5V$ or $V_I > V_{CC} + 0.5V$ | –20 | 20 | mA |
| | DC switch current(2) | $V_I < V_{EE} – 0.5V$ or $V_I > V_{CC} + 0.5V$ | -25 | 25 | mA |
| $I_{CC}$ | DC $V_{CC}$ or ground current | | –50 | 50 | mA |
| $I_{EE}$ | DC $V_{EE}$ current | | –20 | | mA |
| $V_{SEL}$ or $V_{EN}$ | Logic control input pin voltage ($\overline{EN}$, Ax, SELx) | | –0.5 | 30 | V |
| $T_{JMAX}$ | Maximum junction temperature | | | 150 | °C |
| $T_{LMAX}$ | Maximum lead temperature | Soldering 10s | | 300 | °C |
| $T_{stg}$ | Storage temperature | | –65 | 150 | °C |

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) All voltages are with respect to ground, unless otherwise specified.

## 5.2 ESD Ratings

| | | | VALUE | UNIT |
|---|---|---|---|---|
| $V_{(ESD)}$ | Electrostatic discharge | Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins(1) | ±500 | V |
| | | Charged device model (CDM), per JEDEC specification JESD22-C101, all pins(2) | ±200 | |

(1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.


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## 5.3 Thermal Information

| THERMAL METRIC[1] | | CD74HC4051 | | | UNIT |
|---|---|---|---|---|---|
| | | N (PDIP) | NS (SO) | PW (TSSOP) | |
| | | 16 PINS | 16 PINS | 16 PINS | |
| $R_{\theta JA}$ | Junction-to-ambient thermal resistance | 77.3 | 99.3 | 116.5 | °C/W |
| $R_{\theta JC(top)}$ | Junction-to-case (top) thermal resistance | 56.2 | 59.6 | 51.9 | °C/W |
| $R_{\theta JB}$ | Junction-to-board thermal resistance | 52.6 | 65.7 | 73.9 | °C/W |
| $\Psi_{JT}$ | Junction-to-top characterization parameter | 33.7 | 21.5 | 4.7 | °C/W |
| $\Psi_{JB}$ | Junction-to-board characterization parameter | 52.1 | 65.1 | 73.2 | °C/W |

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application note.

## 5.4 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)[1]

| | | | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|---|
| $V_{CC}$ | Supply voltage range ($T_A$ = full package temperature range)[2] | CD54 and 74HC types | 2 | | 6 | V |
| | | CD54 and 74HCT types | 4.5 | | 5.5 | |
| $V_{CC} – V_{EE}$ | Supply voltage range ($T_A$ = full package temperature range) | CD54 and 74HC types, CD54 and 74HCT types | 2 | | 10 | V |
| $V_{EE}$ | Supply voltage range ($T_A$ = full package temperature range)[3] | CD54 and 74HC types, CD54 and 74HCT types | 0 | | –6 | V |
| $V_I$ | DC input control voltage | | 0 | | $V_{CC}$ | V |
| $V_{IS}$ | Analog switch I/O voltage | | $V_{EE}$ | | $V_{CC}$ | V |
| $T_A$ | Ambient temperature | | –55 | | 125 | °C |
| $t_r$ , $t_f$ | Input rise and fall times | 2V | 0 | | 1000 | ns |
| | | 4.5V | 0 | | 500 | |
| | | 6V | 0 | | 400 | |

(1) For maximum reliability, nominal operating conditions must be selected so that operation is always within the ranges specified in the *Recommended Operating Conditions* table.
(2) All voltages referenced to GND unless otherwise specified.
(3) In certain applications, the external load resistor current may include both $V_{CC}$ and signal line components. To avoid drawing $V_{CC}$ current when switch current flows into the transmission gate inputs, the voltage drop across the bidirectional switch must not exceed 0.6V (calculated from $r_{ON}$ values shown in *Electrical Characteristics HC* and *Electrical Characteristics HCT* tables). No $V_{CC}$ current will flow through $R_L$ if the switch current flows into terminal 3 on the HC and HCT40511; terminals 3 and 13 on the HC and HCT4052; terminals 4, 14, and 15 on the HC and HCT4053.

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## 5.5 Electrical Characteristics: HC Devices

Over operating free-air temperature range, $V_{SUPPLY}$ = ±5V, and $R_L$ = 100Ω, (unless otherwise noted)

| PARAMETER | TEST CONDITIONS | | | | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| **CD74HC405x** | | | | | | | | | |
| | $V_{IS}$ (V) | $V_I$ (V) | $V_{EE}$ (V) | $V_{CC}$ (V) | $T_A$ | | | | |
| Input High Voltage, $V_{IH}$, Min | | | | 2 | 25°C | 1.5 | | | V |
| | | | | | –40°C to +85°C | 1.5 | | | |
| | | | | | –55°C to +125°C | 1.5 | | | |
| | | | | 4.5 | 25°C | 3.15 | | | |
| | | | | | –40°C to +85°C | 3.15 | | | |
| | | | | | –55°C to +125°C | 3.15 | | | |
| | | | | 6 | 25°C | 4.2 | | | |
| | | | | | –40°C to +85°C | 4.2 | | | |
| | | | | | –55°C to +125°C | 4.2 | | | |
| Input Low Voltage, $V_{IL}$, Max | | | | 2 | 25°C | | | 0.5 | V |
| | | | | | –40°C to +85°C | | | 0.5 | |
| | | | | | –55°C to +125°C | | | 0.5 | |
| | | | | 4.5 | 25°C | | | 1.35 | |
| | | | | | –40°C to +85°C | | | 1.35 | |
| | | | | | –55°C to +125°C | | | 1.35 | |
| | | | | 6 | 25°C | | | 1.8 | |
| | | | | | –40°C to +85°C | | | 1.8 | |
| | | | | | –55°C to +125°C | | | 1.8 | |


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## 5.5 Electrical Characteristics: HC Devices (continued)

Over operating free-air temperature range, $V_{SUPPLY}$ = ±5V, and $R_L$ = 100Ω, (unless otherwise noted)

| PARAMETER | TEST CONDITIONS | | | | | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|
| $r_{ON}$ ON resistance | $I_O$ = 1mA | $V_{CC}$ or $V_{EE}$ | $V_{IL}$ or $V_{IH}$ | 0 | 4.5 | 25°C | | 70 | 160 | Ω |
| | | | | | | –40°C to +85°C | | | 200 | |
| | | | | | | –55°C to +125°C | | | 240 | |
| | | | | 0 | 6 | 25°C | | 60 | 140 | |
| | | | | | | –40°C to +85°C | | | 175 | |
| | | | | | | –55°C to +125°C | | | 210 | |
| | | | | -4.5 | 4.5 | 25°C | | 40 | 120 | |
| | | | | | | –40°C to +85°C | | | 150 | |
| | | | | | | –55°C to +125°C | | | 180 | |
| | | $V_{CC}$ to $V_{EE}$ | | 0 | 4.5 | 25°C | | 90 | 180 | Ω |
| | | | | | | –40°C to +85°C | | | 225 | |
| | | | | | | –55°C to +125°C | | | 270 | |
| | | | | 0 | 6 | 25°C | | 80 | 160 | |
| | | | | | | –40°C to +85°C | | | 200 | |
| | | | | | | –55°C to +125°C | | | 240 | |
| | | | | -4.5 | 4.5 | 25°C | | 45 | 130 | |
| | | | | | | –40°C to +85°C | | | 162 | |
| | | | | | | –55°C to +125°C | | | 195 | |
| $\Delta r_{ON}$ Maximum ON resistance between any two channels | | | | 0 | 4.5 | 25°C | | 10 | | Ω |
| | | | | 0 | 6 | 25°C | | 8.5 | | |
| | | | | -4.5 | 4.5 | 25°C | | 5 | | |

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## 5.5 Electrical Characteristics: HC Devices (continued)

Over operating free-air temperature range, $V_{SUPPLY}$ = ±5V, and $R_L$ = 100Ω, (unless otherwise noted)

| PARAMETER | TEST CONDITIONS | | | | | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|
| $I_{IZ}$ Switch ON/OFF leakage current | 1 and 2 channels | For switch OFF: When $V_{IS} = V_{CC}$, $V_{OS} = V_{EE}$; When $V_{IS} = V_{EE}$, $V_{OS} = V_{CC}$, For switch ON: All applicable combinations of $V_{IS}$ and $V_{OS}$ voltage levels | $V_{IL}$ or $V_{IH}$ | 0 | 6 | 25°C | | | ±0.1 | µA |
| | | | | | | –55°C to 85°C | | | ±1 | |
| | | | | | | –55°C to 125°C | | | ±1 | |
| | 4053 | | | -5 | 5 | 25°C | | | ±0.1 | |
| | | | | | | –55°C to 85°C | | | ±1 | |
| | | | | | | –55°C to 125°C | | | ±1 | |
| | 4 channels | | | 0 | 6 | 25°C | | | ±0.1 | |
| | | | | | | –55°C to 85°C | | | ±1 | |
| | | | | | | –55°C to 125°C | | | ±1 | |
| | 4052 | | | -5 | 5 | 25°C | | | ±0.2 | |
| | | | | | | –55°C to 85°C | | | ±2 | |
| | | | | | | –55°C to 125°C | | | ±2 | |
| | 8 channels | | | 0 | 6 | 25°C | | | ±0.2 | |
| | | | | | | –55°C to 85°C | | | ±2 | |
| | | | | | | –55°C to 125°C | | | ±2 | |
| | 4051 | | | -5 | 5 | 25°C | | | ±0.4 | |
| | | | | | | –55°C to 85°C | | | ±4 | |
| | | | | | | –55°C to 125°C | | | ±4 | |
| $I_{IL}$ Control input leakage current | | | $V_{CC}$ or GND | 0 | 6 | 25°C | | | ±0.1 | µA |
| | | | | | | –55°C to 85°C | | | ±1 | |
| | | | | | | –55°C to 125°C | | | ±1 | |
| Quiescent Device Current, $I_{CC}$ Max | $I_O$ = 0 | When $V_{IS} = V_{EE}$, $V_{OS} = V_{CC}$ | $V_{CC}$ or GND | 0 | 6 | 25°C | | | 12 | µA |
| | | | | | | –55°C to 85°C | | | 80 | |
| | | | | | | –55°C to 125°C | | | 160 | |
| | | When $V_{IS} = V_{CC}$, $V_{OS} = V_{EE}$ | | -5 | 5 | 25°C | | | 32 | |
| | | | | | | –55°C to 85°C | | | 160 | |
| | | | | | | –55°C to 125°C | | | 320 | |


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## 5.6 Electrical Characteristics: HCT Devices

Over operating free-air temperature range, $V_{SUPPLY}$ = ±5V, and $R_L$ = 100Ω, (unless otherwise noted)(1)

| PARAMETER | | TEST CONDITIONS | | | | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|
| **CD74HCT405x** | | | | | | | | | | |
| | | **$V_{IS}$ (V)** | **$V_I$ (V)** | **$V_{EE}$ (V)** | **$V_{CC}$ (V)** | **$T_A$** | | | | |
| Input High Voltage, $V_{IH}$, Min | | | | | 4.5 to 5.5 | 25°C | 2 | | | V |
| | | | | | | –40°C to +85°C | 2 | | | |
| | | | | | | –55°C to +125°C | 2 | | | |
| Input Low Voltage, $V_{IL}$, Max | | | | | 4.5 to 5.5 | 25°C | | | 0.8 | V |
| | | | | | | –40°C to +85°C | | | 0.8 | |
| | | | | | | –55°C to +125°C | | | 0.8 | |
| $r_{ON}$ ON resistance | $I_O$ = 1mA | $V_{CC}$ or $V_{EE}$ | $V_{IL}$ or $V_{IH}$ | 0 | 4.5 | 25°C | | 70 | 160 | Ω |
| | | | | | | –40°C to +85°C | | | 200 | |
| | | | | | | –55°C to +125°C | | | 240 | |
| | | | | -4.5 | 4.5 | 25°C | | 40 | 120 | |
| | | | | | | –40°C to +85°C | | | 150 | |
| | | | | | | –55°C to +125°C | | | 180 | |
| | | $V_{CC}$ to $V_{EE}$ | | 0 | 4.5 | 25°C | | 90 | 180 | |
| | | | | | | –40°C to +85°C | | | 225 | |
| | | | | | | –55°C to +125°C | | | 270 | |
| | | | | -4.5 | 4.5 | 25°C | | 45 | 130 | Ω |
| | | | | | | –40°C to +85°C | | | 162 | |
| | | | | | | –55°C to +125°C | | | 195 | |
| $\Delta r_{ON}$ Maximum ON resistance between any two channels | | | | 0 | 4.5 | 25°C | | 10 | | Ω |
| | | | | -4.5 | 4.5 | 25°C | | 5 | | |

## 5.6 Electrical Characteristics: HCT Devices (continued)

Over operating free-air temperature range, $V_{SUPPLY}$ = ±5V, and $R_L$ = 100Ω, (unless otherwise noted)(1)

| PARAMETER | TEST CONDITIONS | | | | | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|
| $I_{IZ}$ Switch ON/OFF leakage current | 1 and 2 channels | For switch OFF: When $V_{IS}$ = $V_{CC}$, $V_{OS}$ = $V_{EE}$; When $V_{IS}$ = $V_{EE}$, $V_{OS}$ = $V_{CC}$, For switch ON: All applicable combinations of $V_{IS}$ and $V_{OS}$ voltage levels | $V_{IL}$ or $V_{IH}$ | 0 | 6 | 25°C | | | ±0.1 | µA |
| | | | | | | –55°C to 85°C | | | ±1 | |
| | | | | | | –55°C to 125°C | | | ±1 | |
| | 4053 | | | -5 | 5 | 25°C | | | ±0.1 | |
| | | | | | | –55°C to 85°C | | | ±1 | |
| | | | | | | –55°C to 125°C | | | ±1 | |
| | 4 channels | | | 0 | 6 | 25°C | | | ±0.1 | |
| | | | | | | –55°C to 85°C | | | ±1 | |
| | | | | | | –55°C to 125°C | | | ±1 | |
| | 4052 | | | -5 | 5 | 25°C | | | ±0.2 | |
| | | | | | | –55°C to 85°C | | | ±2 | |
| | | | | | | –55°C to 125°C | | | ±2 | |
| | 8 channels | | | 0 | 6 | 25°C | | | ±0.2 | |
| | | | | | | –55°C to 85°C | | | ±2 | |
| | | | | | | –55°C to 125°C | | | ±2 | |
| | 4051 | | | -5 | 5 | 25°C | | | ±0.4 | |
| | | | | | | –55°C to 85°C | | | ±4 | |
| | | | | | | –55°C to 125°C | | | ±4 | |
| $I_{IL}$ Control input leakage current | | | See(1) | 0 | 5.5 | 25°C | | | ±0.1 | µA |
| | | | | | | –55°C to 85°C | | | ±1 | |
| | | | | | | –55°C to 125°C | | | ±1 | |
| Quiescent Device Current, $I_{CC}$ Max | $I_O$ = 0 | When $V_{IS}$ = $V_{EE}$, $V_{OS}$ = $V_{CC}$ | $V_{CC}$ or GND | 0 | 5.5 | 25°C | | | 12 | µA |
| | | | | | | –55°C to 85°C | | | 80 | |
| | | | | | | –55°C to 125°C | | | 160 | |
| | | When $V_{IS}$ = $V_{CC}$, $V_{OS}$ = $V_{EE}$ | | -4.5 | 5.5 | 25°C | | | 32 | |
| | | | | | | –55°C to 85°C | | | 160 | |
| | | | | | | –55°C to 125°C | | | 320 | |
| $\Delta I_{CC}$ Additional quiescent device current per input pin: 1 unit load(2) | | ΔICC | $V_{CC}$ - 2.1 | | 4.5 to 5.5 | 25°C | | 100 | 360 | µA |
| | | | | | | –55°C to 85°C | | | 450 | |
| | | | | | | –55°C to 125°C | | | 490 | |

(1) Any voltage between $V_{CC}$ and GND.
(2) For dual-supply systems, theoretical worse-case ($V_I$ = 2.4V, $V_{CC}$ = 5.5V) specification is 1.8mA.


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## 5.7 Switching Characteristics, VCC = 5V

$V_{CC}$ = 5V, $T_A$ = 25°C, input $t_r$,$t_f$ = 6ns

| Parameter | | Test Conditions | | $C_L$ (pF) | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| $t_{PHL}$, $t_{PLH}$ | Supply voltage range ($T_A$ = full package temperature range) | Switch IN to OUT | CDx4HC4051 | 15 | | 4 | | ns |
| | | | CDx4HCT4051 | | | 4 | | |
| | | | CDx4HC4052 | | | 4 | | |
| | | | CDx4HCT4052 | | | 4 | | |
| | | | CDx4HC4053 | | | 4 | | |
| | | | CDx4HCT4053 | | | 4 | | |
| $t_{PHZ}$, $t_{PLZ}$ | | Switch turn-off (S or E) | CDx4HC4051 | 15 | | 27 | | |
| | | | CDx4HCT4051 | | | 35 | | |
| | | | CDx4HC4052 | | | 33 | | |
| | | | CDx4HCT4052 | | | 33 | | |
| | | | CDx4HC4053 | | | 30 | | |
| | | | CDx4HCT4053 | | | 35 | | |
| $t_{PZH}$, $t_{PZL}$ | | Switch turn-on (S or E) | CDx4HC4051 | 15 | | 19 | | |
| | | | CDx4HCT4051 | | | 23 | | |
| | | | CDx4HC4052 | | | 27 | | |
| | | | CDx4HCT4052 | | | 29 | | |
| | | | CDx4HC4053 | | | 18 | | |
| | | | CDx4HCT4053 | | | 28 | | |
| $C_{PD}$ Power dissipation capacitance[(1)] | | | CDx4HC4051 | | | 50 | | pF |
| | | | CDx4HCT4051 | | | 52 | | |
| | | | CDx4HC4052 | | | 74 | | |
| | | | CDx4HCT4052 | | | 76 | | |
| | | | CDx4HC4053 | | | 38 | | |
| | | | CDx4HCT4053 | | | 42 | | |

(1) $C_{PD}$ is used to determine the dynamic power consumption, per package. $P_D = C_{PD} v_{CC}{}^2 f_i + \Sigma (C_L + C_S) V_{CC}{}^2 f_O$, $f_O$= output frequency, $f_I$ = input frequency, $C_L$ = output load capacitance, $C_S$ = switch capacitance, $V_{CC}$ = supply voltage

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## 5.8 Switching Characteristics, CL = 50pF

$C_L$ = 50pF, input $t_r$, $t_f$ = 6ns

| Parameter | | $V_{EE}$ (V) | $V_{CC}$ (V) | Test Conditions | | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| $t_{PHL}$, $t_{PLH}$ Propagation delay, switch in to out | | 0 | 2 | $T_A$ = 25°C | HC | | | 60 | ns |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 75 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 90 | |
| | | 0 | 4.5 | $T_A$ = 25°C | HC, HCT | | | 12 | |
| | | | | $T_A$ = –40°C to +85°C | HC, HCT | | | 15 | |
| | | | | $T_A$ = –55°C to +125°C | HC, HCT | | | 18 | |
| | | 0 | 6 | $T_A$ = 25°C | HC | | | 10 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 13 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 15 | |
| | | -4.5 | 4.5 | $T_A$ = 25°C | HC, HCT | | | 8 | |
| | | | | $T_A$ = –40°C to +85°C | HC, HCT | | | 10 | |
| | | | | $T_A$ = –55°C to +125°C | HC, HCT | | | 12 | |
| $t_{PHZ}$, $t_{PLZ}$ Maximum switch turn OFF delay from S or E to switch output | 4051 | 0 | 2 | $T_A$ = 25°C | HC | | | 250 | ns |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 340 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 400 | |
| | | 0 | 4.5 | $T_A$ = 25°C | HC, HCT | | | 50 | |
| | | | | $T_A$ = –40°C to +85°C | HC, HCT | | | 56 | |
| | | | | $T_A$ = –55°C to +125°C | HC, HCT | | | 68 | |
| | | 0 | 6 | $T_A$ = 25°C | HC | | | 44 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 50 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 57 | |
| | | -4.5 | 4.5 | $T_A$ = 25°C | HC, HCT | | | 44 | |
| | | | | $T_A$ = –40°C to +85°C | HC, HCT | | | 50 | |
| | | | | $T_A$ = –55°C to +125°C | HC, HCT | | | 55 | |


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## 5.8 Switching Characteristics, CL = 50pF (continued)

$C_L$ = 50pF, input $t_r$, $t_f$ = 6ns

| Parameter | | $V_{EE}$ (V) | $V_{CC}$ (V) | Test Conditions | | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| $t_{PHZ}$, $t_{PLZ}$ Maximum switch turn OFF delay from S or E to switch output | 4052 | 0 | 2 | $T_A$ = 25°C | HC | | | 250 | ns |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 340 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 400 | |
| | | 0 | 4.5 | $T_A$ = 25°C | HC, HCT | | | 50 | |
| | | | | $T_A$ = –40°C to +85°C | HC, HCT | | | 63 | |
| | | | | $T_A$ = –55°C to +125°C | HC, HCT | | | 75 | |
| | | 0 | 6 | $T_A$ = 25°C | HC | | | 45 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 54 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 65 | |
| | | -4.5 | 4.5 | $T_A$ = 25°C | HC | | | 45 | |
| | | | | | HCT | | | 45 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 48 | |
| | | | | | HCT | | | 50 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 57 | |
| | | | | | HCT | | | 57 | |
| $t_{PHZ}$, $t_{PLZ}$ Maximum switch turn OFF delay from S or E to switch output | 4053 | 0 | 2 | $T_A$ = 25°C | HC | | | 250 | ns |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 340 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 400 | |
| | | 0 | 4.5 | $T_A$ = 25°C | HC | | | 45 | |
| | | | | | HCT | | | 50 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 53 | |
| | | | | | HCT | | | 53 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 63 | |
| | | | | | HCT | | | 66 | |
| | | 0 | 6 | $T_A$ = 25°C | HC | | | 45 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 50 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 55 | |
| | | -4.5 | 4.5 | $T_A$ = 25°C | HC | | | 45 | |
| | | | | | HCT | | | 45 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 50 | |
| | | | | | HCT | | | 50 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 55 | |
| | | | | | HCT | | | 55 | |

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## 5.8 Switching Characteristics, CL = 50pF (continued)

$C_L$ = 50pF, input $t_r$, $t_f$ = 6ns

| Parameter | | $V_{EE}$ (V) | $V_{CC}$ (V) | Test Conditions | | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| $t_{PZL}$, $t_{PZH}$ Maximum switch turn ON delay from S or E to switch output | 4051 | 0 | 2 | $T_A$ = 25°C | HC | | | 325 | ns |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 405 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 490 | |
| | | 0 | 4.5 | $T_A$ = 25°C | HC | | | 45 | |
| | | | | | HCT | | | 55 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 56 | |
| | | | | | HCT | | | 69 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 68 | |
| | | | | | HCT | | | 83 | |
| | | 0 | 6 | $T_A$ = 25°C | HC | | | 38 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 48 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 57 | |
| | | -4.5 | 4.5 | $T_A$ = 25°C | HC | | | 36 | |
| | | | | | HCT | | | 48 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 40 | |
| | | | | | HCT | | | 55 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 48 | |
| | | | | | HCT | | | 60 | |
| $t_{PZL}$, $t_{PZH}$ Maximum switch turn ON delay from S or E to switch output | 4052 | 0 | 2 | $T_A$ = 25°C | HC | | | 325 | ns |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 405 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 490 | |
| | | 0 | 4.5 | $T_A$ = 25°C | HC | | | 65 | |
| | | | | | HCT | | | 70 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 81 | |
| | | | | | HCT | | | 68 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 98 | |
| | | | | | HCT | | | 105 | |
| | | 0 | 6 | $T_A$ = 25°C | HC | | | 55 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 69 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 83 | |
| | | -4.5 | 4.5 | $T_A$ = 25°C | HC | | | 46 | |
| | | | | | HCT | | | 48 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 58 | |
| | | | | | HCT | | | 60 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 69 | |
| | | | | | HCT | | | 72 | |


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## 5.8 Switching Characteristics, CL = 50pF (continued)

$C_L$ = 50pF, input $t_r$, $t_f$ = 6ns

| Parameter | | $V_{EE}$ (V) | $V_{CC}$ (V) | Test Conditions | | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| $t_{PZL}$, $t_{PZH}$ Maximum switch turn ON delay from S or E to switch output | 4053 | 0 | 2 | $T_A$ = 25°C | HC | | | 325 | ns |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 405 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 490 | |
| | | 0 | 4.5 | $T_A$ = 25°C | HC | | | 44 | |
| | | | | | HCT | | | 48 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 55 | |
| | | | | | HCT | | | 60 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 66 | |
| | | | | | HCT | | | 72 | |
| | | 0 | 6 | $T_A$ = 25°C | HC | | | 37 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 47 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 56 | |
| | | -4.5 | 4.5 | $T_A$ = 25°C | HC | | | 40 | |
| | | | | | HCT | | | 48 | |
| | | | | $T_A$ = –40°C to +85°C | HC | | | 45 | |
| | | | | | HCT | | | 55 | |
| | | | | $T_A$ = –55°C to +125°C | HC | | | 47 | |
| | | | | | HCT | | | 60 | |
| $C_I$ Input (control) capacitance | | | | $T_A$ = 25°C | HC, HCT | | | 10 | pF |
| | | | | $T_A$ = –40°C to +85°C | HC, HCT | | | 10 | |
| | | | | $T_A$ = –55°C to +125°C | HC, HCT | | | 10 | |

## 5.9 Analog Channel Specifications

Typical values at $T_A$ = 25°C

| Parameter | Test Conditions | HC, HCT TYPES | $V_{EE}$ (V) | $V_{CC}$ (V) | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| $C_I$ Switch input capacitance | | All | | | | 5 | | pF |
| $C_{COM}$ Common output capacitance | | 4051 | | | | 25 | | pF |
| | | 4052 | | | | 12 | | |
| | | 4053 | | | | 8 | | |
| $f_{MAX}$ Minimum switch frequency response at –3 dB | See note(1) and (2) | 4051 | -2.25 | 2.25 | | 145 | | MHz |
| | | 4052 | -2.25 | 2.25 | | 165 | | |
| | | 4053 | -2.25 | 2.25 | | 200 | | |
| | | 4051 | -4.5 | 4.5 | | 180 | | |
| | | 4052 | -4.5 | 4.5 | | 185 | | |
| | | 4053 | -4.5 | 4.5 | | 200 | | |
| THD Sine-wave distortion | | All | -2.25 | 2.25 | | 0.035 | | % |
| | | All | -4.5 | 4.5 | | 0.018 | | |

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## 5.9 Analog Channel Specifications (continued)

Typical values at $T_A$ = 25°C

| Parameter | Test Conditions | HC, HCT TYPES | $V_{EE}$ (V) | $V_{CC}$ (V) | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| Switch OFF signal feedthrough | See note(2) and (3) | 4051 | -2.25 | 2.25 | | -73 | | dB |
| | | 4052 | -2.25 | 2.25 | | -65 | | |
| | | 4053 | -2.25 | 2.25 | | -64 | | |
| | | 4051 | -4.5 | 4.5 | | -75 | | |
| | | 4052 | -4.5 | 4.5 | | -67 | | |
| | | 4053 | -4.5 | 4.5 | | -66 | | |

(1) Adjust input voltage to obtain 0dBm at $V_{OS}$ for $f_{IN}$ = 1MHz.
(2) $V_{is}$ is centered at ($V_{CC}$ - $V_{EE}$) / 2.
(3) Adjust input for 0dBm.


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## 5.10 Typical Characteristics

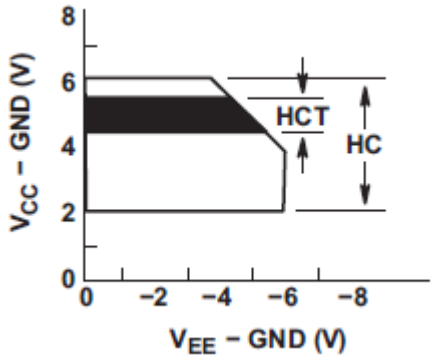



Figure 5-1. Recommended Operating Area as a Function of ($V_{CC}$ - $V_{EE}$)

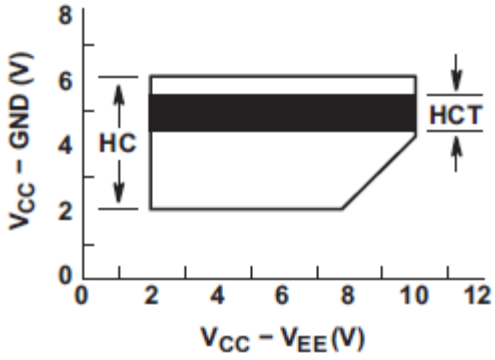



Figure 5-2. Recommended Operating Area as a Function of ($V_{CC}$ - GND)

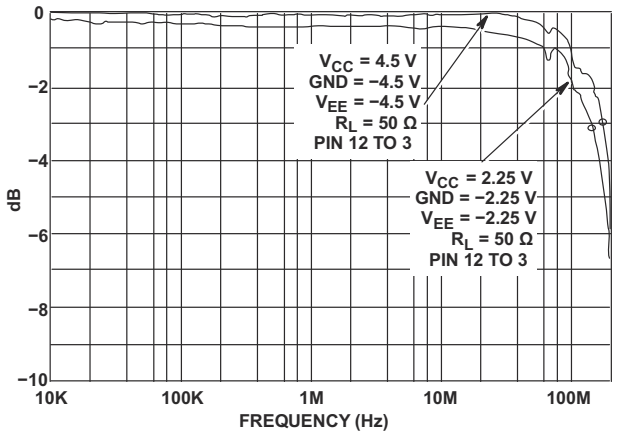



Figure 5-3. Channel ON Bandwidth (HC and HCT4051)

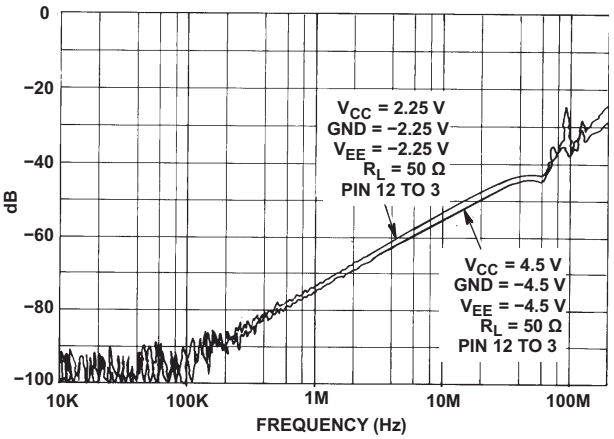



Figure 5-4. Channel OFF Feedthrough (HC and HCT4051)

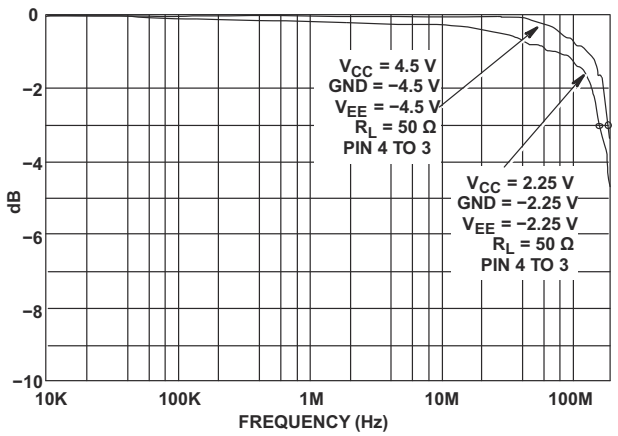



Figure 5-5. Channel ON Bandwidth (HC and HCT4052)

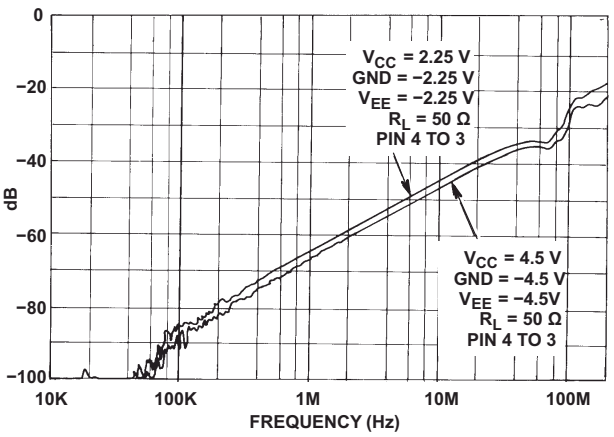



Figure 5-6. Channel OFF Feedthrough (HC and HCT4052)

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## 5.10 Typical Characteristics (continued)

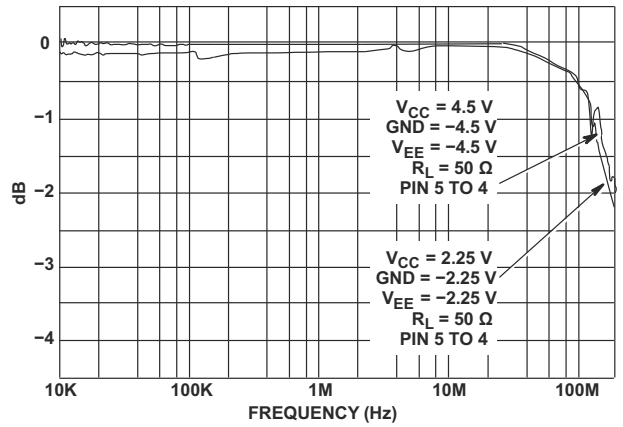



Figure 5-7. Channel ON Bandwidth (HC and HCT4053)

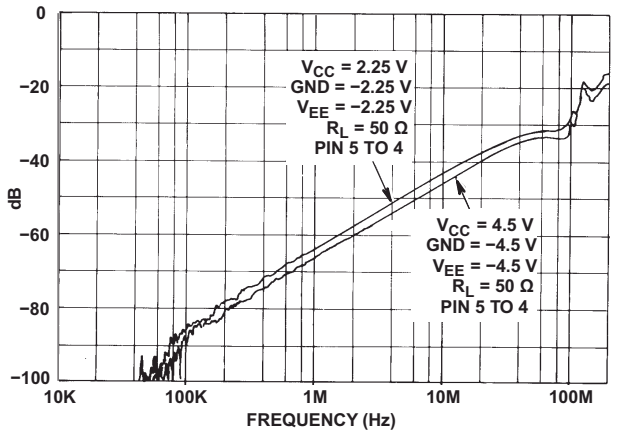
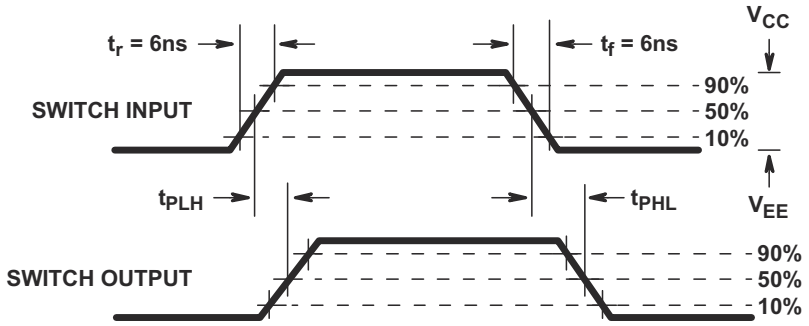
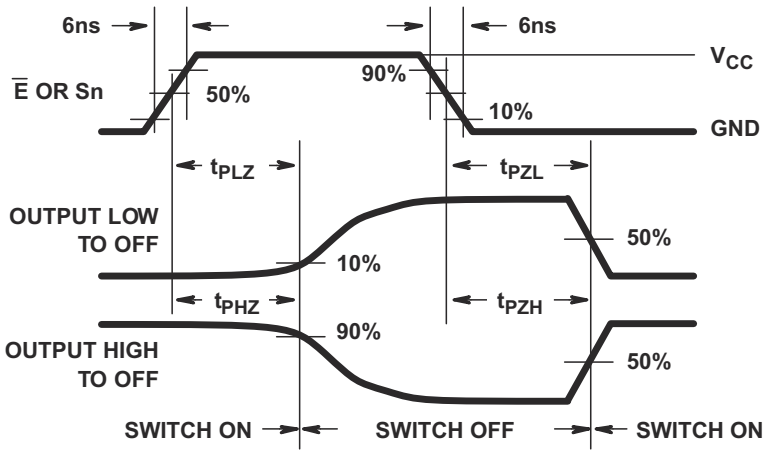



Figure 5-8. Channel OFF Feedthrough (HC and HCT4053)

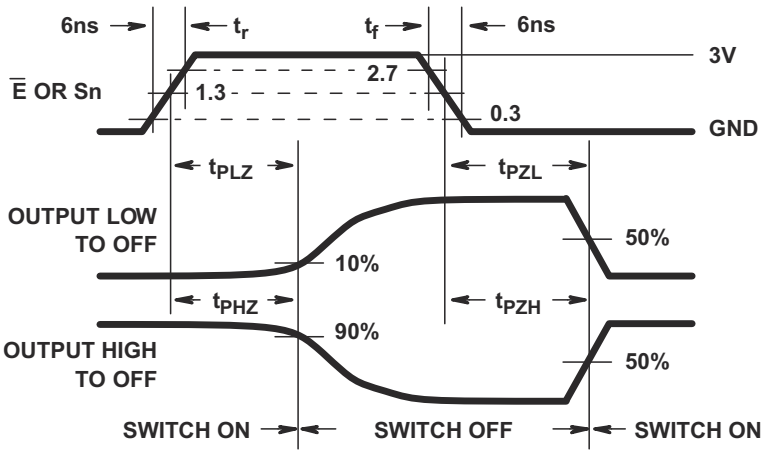
## 6 Parameter Measurement Information




(FIGURE A)




(FIGURE B) HC TYPES




(FIGURE C) HCT TYPES

**Figure 6-1. Switch Propagation Delay, Turn-On, Turn-Off Times**


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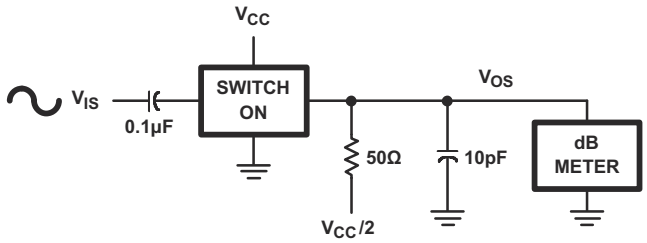
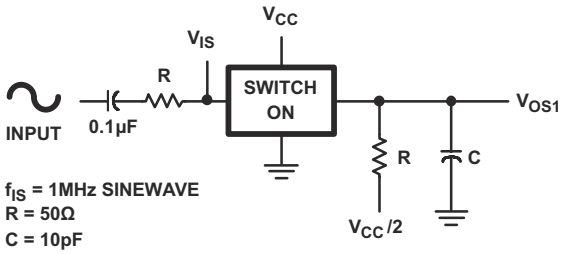



Figure 6-2. Frequency Response Test Circuit



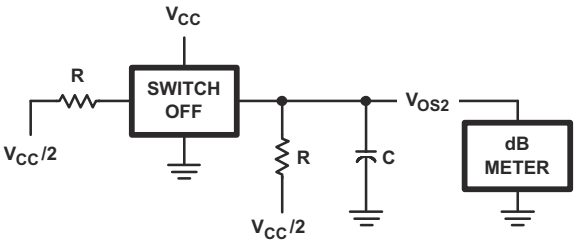




Figure 6-3. Crosstalk Between Two Switches Test Circuit

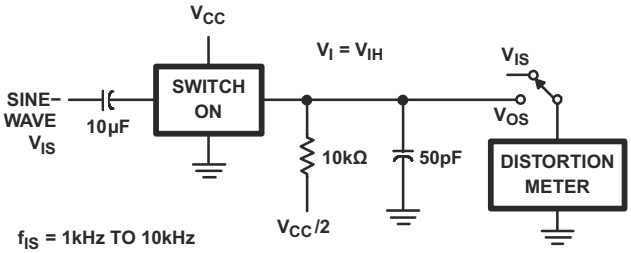



Figure 6-4. ¼Sine-Wave Distortion Test Circuit

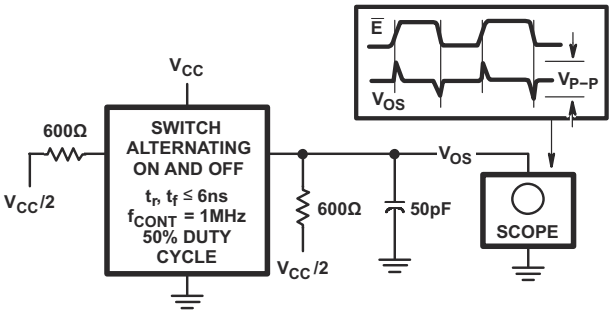



Figure 6-5. Control to Switch Feedthrough Noise Test Circuit

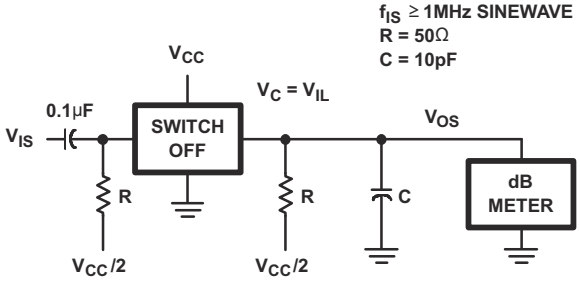



Figure 6-6. Switch OFF Signal Feedthrough

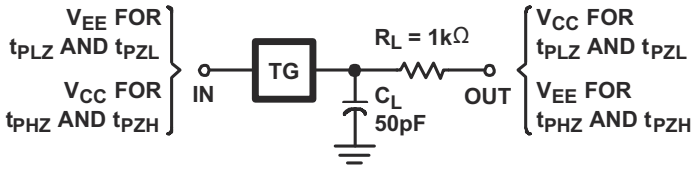



Figure 6-7. Switch ON/OFF Propagation Delay Test Circuit

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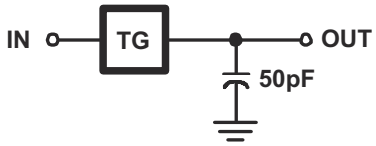



**Figure 6-8. Switch In to Switch Out Propagation Delay Test Circuit**


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# 7 Detailed Description

## 7.1 Overview

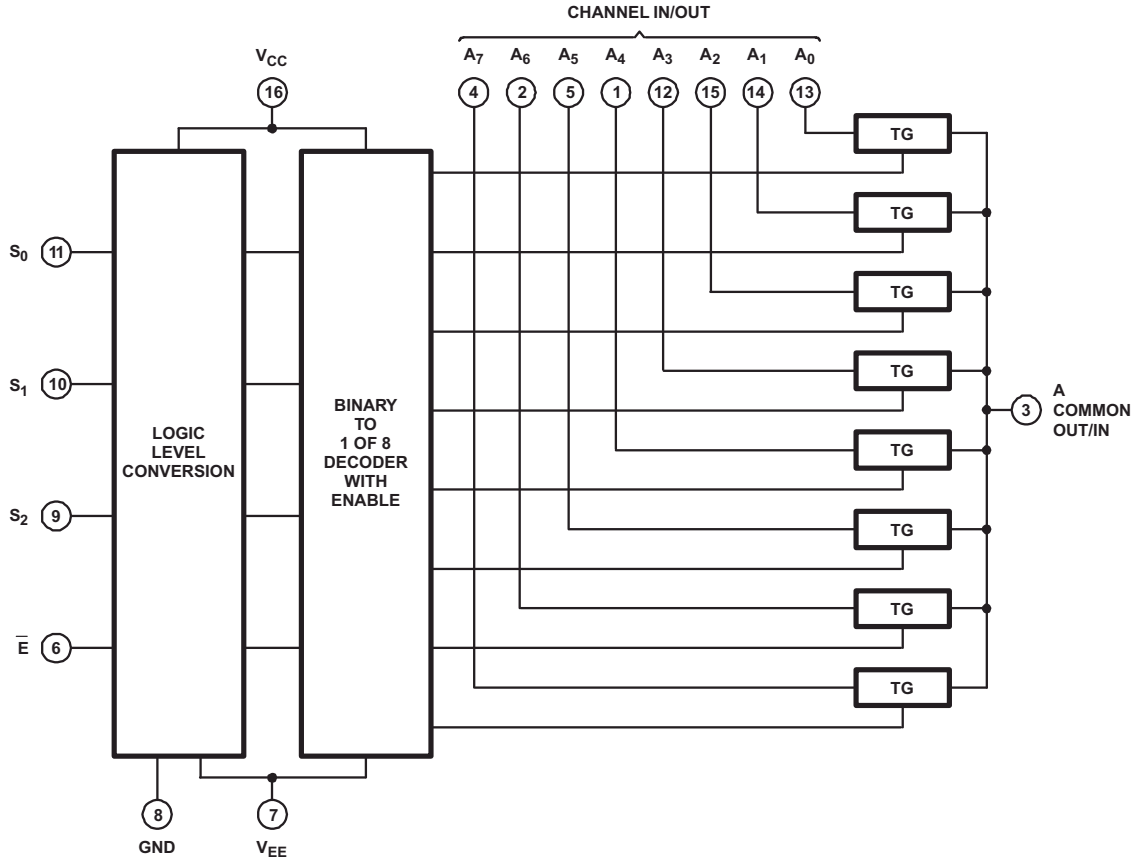
The CDx4HCx4051 devices are a single 8-channel multiplexer having three binary control inputs, $S_0$, $S_1$, and $S_2$ and an $\overline{\text{ENABLE}}$ input. The three binary signals select 1 of 8 channels to be turned on, and connect one of the 8 inputs to the output.

The CDx4HCx4052 devices are a differential 4-channel multiplexer having two binary control inputs, $S_0$ and $S_1$, and an $\overline{\text{ENABLE}}$ input. The two binary input signals select 1 of 4 pairs of channels to be turned on and connect the analog inputs to the outputs.

The CDx4HCx4053 devices are a triple 2-channel multiplexer having three separate digital control inputs, $S_0$, $S_1$, and $S_2$ and an $\overline{\text{ENABLE}}$ input. Each control input selects one of a pair of channels that are connected in a single-pole, double-throw configuration.

When these devices are used as demultiplexers, the CHANNEL IN/OUT terminals are the outputs and the COMMON OUT/IN terminals are the inputs.

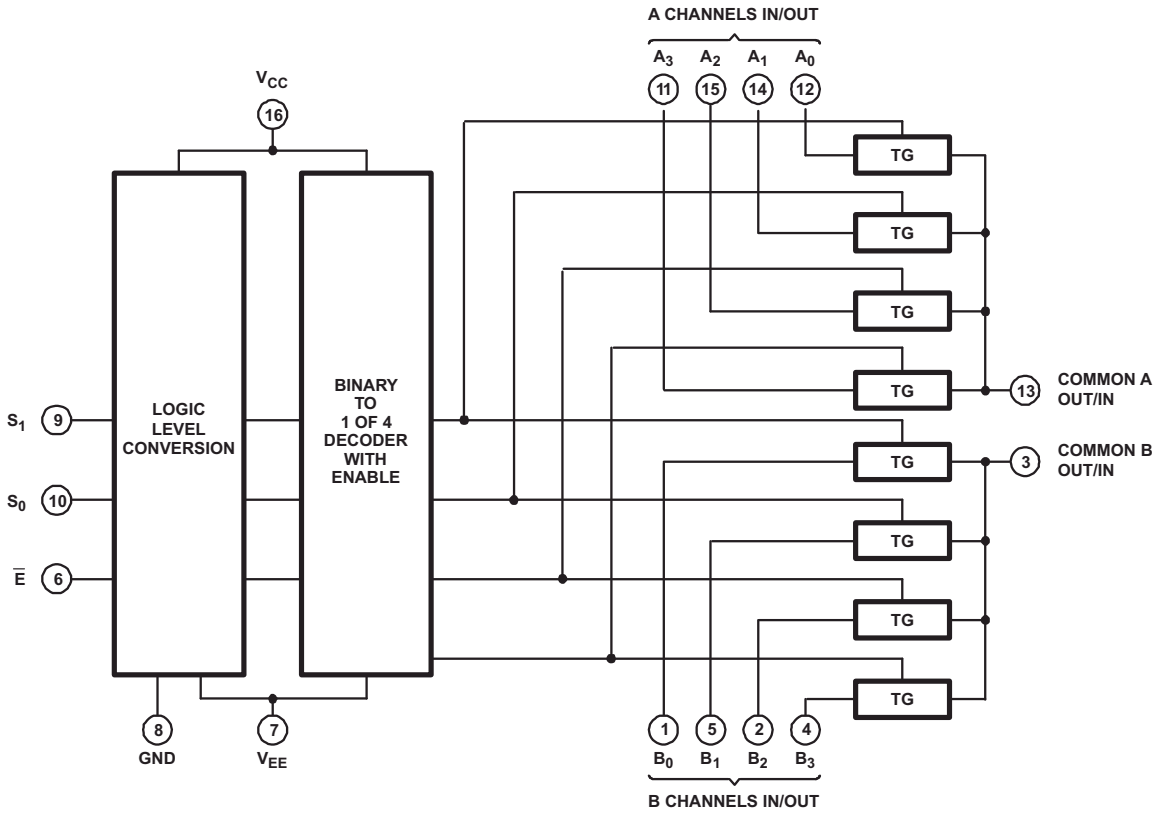
## 7.2 Functional Block Diagrams




All inputs are protected by standard CMOS protection network.

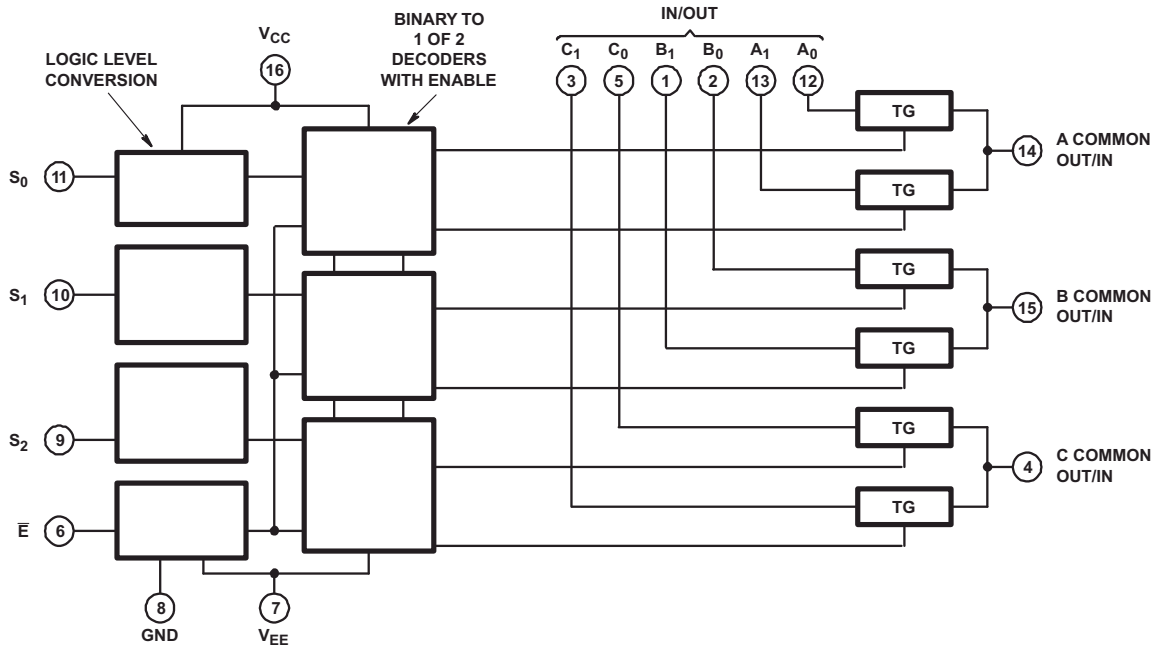
**Figure 7-1. CDx4HCx4051 Functional Block Diagram**

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All inputs are protected by standard CMOS protection network.

**Figure 7-2. CDx4HCx4052 Functional Block Diagram**




All inputs are protected by standard CMOS protection network.

**Figure 7-3. CDx4HCx4053 Functional Block Diagram**


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## 7.3 Feature Description

The CDx4HCx405x line of multiplexers and demultiplexers can accept a wide range of analog signal levels from –5 to +5V. They have low ON resistance, typically 70Ω for $V_{CC} – V_{EE}$ = 4.5V and 40Ω for $V_C – V_{EE}$ = 4.5V, which allows for very little signal loss through the switch.

Binary address decoding on chip makes channel selection easy. When channels are changed, a break-before-make system eliminates channel overlap.

## 7.4 Device Functional Modes

**Table 7-1. CD54HC4051, CD74HC4051, CD54HCT4051, CD74HCT4051 Function Table (1)**

| INPUT STATES | | | | ON CHANNEL |
|---|---|---|---|---|
| ENABLE | $S_2$ | $S_1$ | $S_0$ | |
| L | L | L | L | A0 |
| L | L | L | H | A1 |
| L | L | H | L | A2 |
| L | L | H | H | A3 |
| L | H | L | L | A4 |
| L | H | L | H | A5 |
| L | H | H | L | A6 |
| L | H | H | H | A7 |
| H | X | X | X | None |

(1) X = Don't care

**Table 7-2. CD54HC4052, CD74HC4052, CD54HCT4052, CD74HCT4052 Function Table (1)**

| INPUT STATES | | | ON CHANNELS |
|---|---|---|---|
| ENABLE | $S_1$ | $S_0$ | |
| L | L | L | A0, B0 |
| L | L | H | A1, B1 |
| L | H | L | A2, B2 |
| L | H | H | A3, B3 |
| H | X | X | None |

(1) X = Don't care

**Table 7-3. CD54HC4053, CD74HC4053, CD54HCT4053, CD74HCT4053 Function Table (1)**

| INPUT STATES | | | | ON CHANNELS |
|---|---|---|---|---|
| ENABLE | $S_2$ | $S_1$ | $S_0$ | |
| L | L | L | L | C0, B0, A0 |
| L | L | L | H | C0, B0, A1 |
| L | L | H | L | C0, B1, A0 |
| L | L | H | H | C0, B1, A1 |
| L | H | L | L | C1, B0, A0 |
| L | H | L | H | C1, B0, A1 |
| L | H | H | L | C1, B1, A0 |
| L | H | H | H | C1, B1, A1 |
| H | X | X | X | None |

(1) X = Don't care

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# 8 Application and Implementation

**Note**

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

## 8.1 Application Information

The CDx4HCx405x line of multiplexers and demultiplexers can be used for a wide variety of applications.

## 8.2 Typical Application

One application of the CD74HC4051 device is used in conjunction with a microcontroller to poll a keypad. Figure 8-1 shows the basic schematic for such a polling system. The microcontroller uses the channel-select pins to cycle through the different channels while reading the input to see if a user is pressing any of the keys. This is a very robust setup that allows for simultaneous key presses with very little power consumption. It also uses very few pins on the microcontroller. The down side of polling is that the microcontroller must frequently scan the keys for a press.

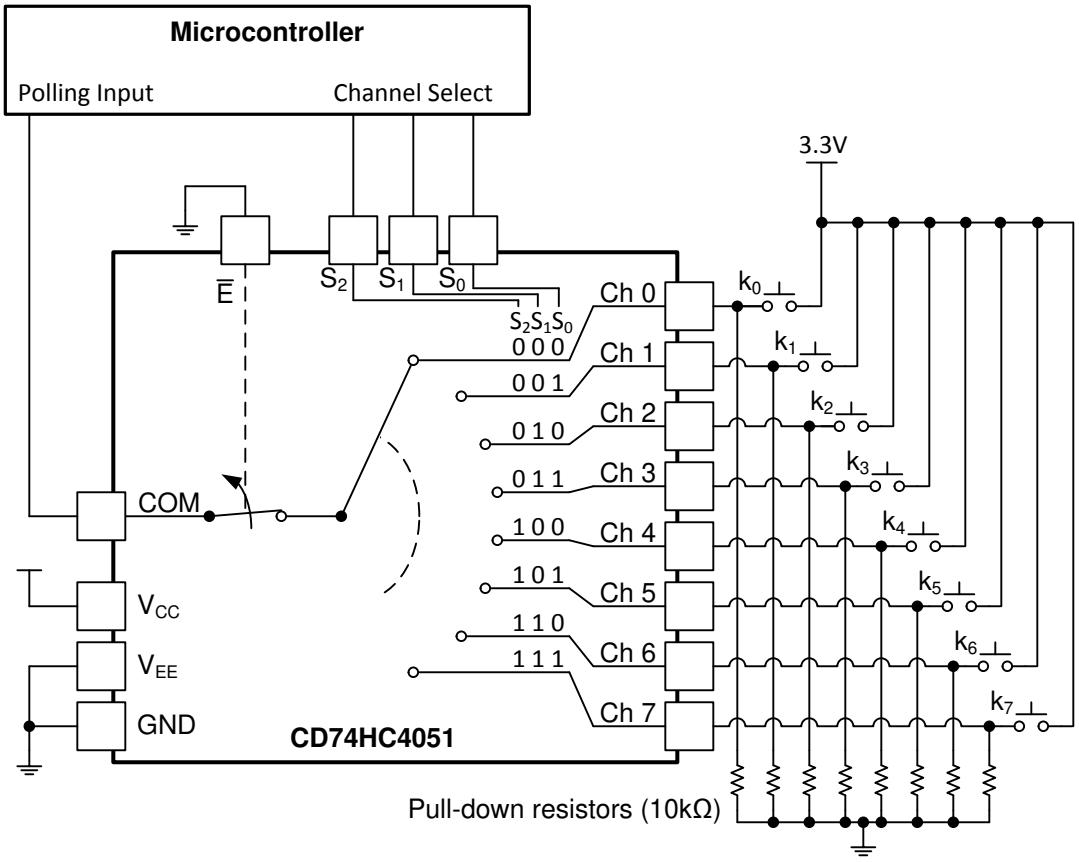



**Figure 8-1. CD74HC4051 Being Used to Help Read Button Presses on a Keypad**

### *8.2.1 Design Requirements*

These devices use CMOS technology and have balanced output drive. Take care to avoid bus contention because it can drive currents that would exceed maximum limits. The high drive will also create fast edges into light loads, so routing and load conditions must be considered to prevent ringing.


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See Table 8-1 for the input loading details.

**Table 8-1. HCT Input Loading Table**

| TYPE | INPUT | UNIT LOADS(1) |
|---|---|---|
| 4051, 4053 | All | 0.5 |
| 4052 | All | 0.4 |

(1) Unit load is $\Delta I_{CC}$ limit specified in *Section 5*, for example, 360mA MAX at 25°C.

### 8.2.2 Detailed Design Procedure

1. Recommended input conditions:
   - For switch time specifications, see propagation delay times in *Section 5.5*.
   - Inputs must not be pushed more than 0.5V above $V_{DD}$ or below $V_{EE}$.
   - For input voltage level specifications for control inputs, see $V_{IH}$ and $V_{IL}$ in *Section 5.5*.
2. Recommended output conditions:
   - Outputs must not be pulled above $V_{DD}$ or below $V_{EE}$.
3. Input and output current consideration:
   - The CDx4HCx405x series of parts do not have internal current-drive circuitry, and thus cannot sink or source current. Any current will be passed through the device.

### 8.2.3 Application Curve

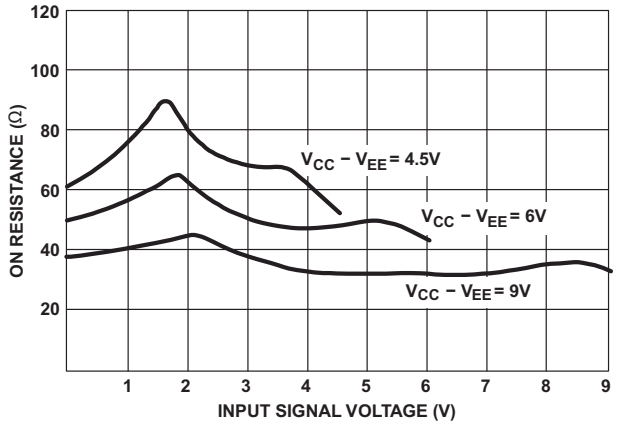



**Figure 8-2. Typical ON Resistance vs Input Signal Voltage**

## 8.3 Power Supply Recommendations

The power supply can be any voltage between the minimum and maximum supply voltage rating located in the *Section 5.5*.

Each $V_{CC}$ terminal must have a good bypass capacitor to prevent power disturbance. For devices with a single supply, a 0.1µF bypass capacitor is recommended. If there are multiple pins labeled $V_{CC}$, then a 0.01µF or 0.022µF capacitor is recommended for each $V_{CC}$ because the $V_{CC}$ pins will be tied together internally. For devices with dual-supply pins operating at different voltages, for example $V_{CC}$ and $V_{DD}$, a 0.1µF bypass capacitor is recommended for each supply pin. It is acceptable to parallel multiple bypass capacitors to reject different frequencies of noise. A 0.1µF and a 1µF capacitor are commonly used in parallel. For best results, the bypass capacitor or capacitors must be installed as close as possible to the power terminal.

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## 8.4 Layout

### *8.4.1 Layout Guidelines*

Reflections and matching are closely related to loop antenna theory, but different enough to warrant their own discussion. When a PCB trace turns a corner at a 90° angle, a reflection can occur. This is primarily due to the change in width of the trace. At the apex of the turn, the trace width is increased to 1.414 times its width. This change in width upsets the transmission line characteristics, especially the distributed capacitance and self-inductance of the trace, thus resulting in the reflection. Not all PCB traces can be straight, so they will have to turn corners. Figure 8-3 shows progressively better techniques of rounding corners. Only the last example (BEST) maintains constant trace width and minimizes reflections.

### *8.4.2 Layout Example*

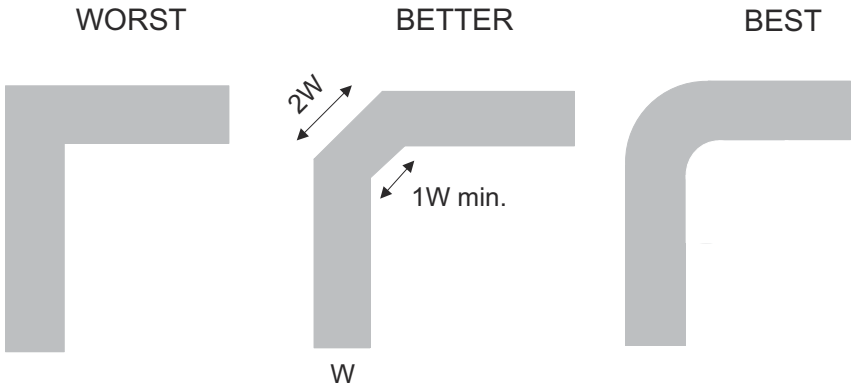



**Figure 8-3. Trace Example**


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# 9 Device and Documentation Support

## 9.1 Documentation Support

### *9.1.1 Related Documentation*

For related documentation, see the following:

- Texas Instruments, *Implications of Slow or Floating CMOS Inputs*

## 9.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

## 9.3 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's Terms of Use.

## 9.4 Trademarks

TI E2E™ is a trademark of Texas Instruments.
All trademarks are the property of their respective owners.

## 9.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

## 9.6 Glossary

| | |
|---|---|
| TI Glossary | This glossary lists and explains terms, acronyms, and definitions. |

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## 10 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| Changes from Revision N (April 2024) to Revision O (January 2026) | Page |
|---|---|
| • Removed "Qualified for automotive applications" feature | 1 |

| Changes from Revision M (May 2019) to Revision N (April 2024) | Page |
|---|---|
| • Changed thermal metrics | 7 |
| • Changed HC ICC at 25°C single/dual supply | 8 |
| • Changed HCT ICC at 25°C single/dual supply | 11 |
| • Changed: tPHZ/tPLZ typicals Switch turn-off (S or E) | 13 |
| • Changed tPHZ/tPLZ maximum switch turn OFF delay from S or E to switch output for 4051/4052/4053 | 14 |
| • Changed tPZL/tPZH maximum switch turn ON delay from S or E to switch output for 4051/4053 | 14 |

| Changes from Revision L (February 2017) to Revision M (May 2019) | Page |
|---|---|
| • Changed *Feature* From: 7Ω Typical To: 70Ω Typical | 1 |

| Changes from Revision K (September 2015) to Revision L (February 2017) | Page |
|---|---|
| • Changed charged device model (CDM) value from: ±1000V to: ±200V | 6 |
| • Added *Receiving Notification of Documentation Updates* section | 6 |

| Changes from Revision J (February 2011) to Revision K (September 2015) | Page |
|---|---|
| • *Added Military Disclaimer to Features* list | 1 |
| • *Removed Ordering Information* table | 1 |
| • Added *Device Information* table, *Pin Functions* table, *ESD Ratings* table, *Thermal Information* table, *Detailed Description* section, *Applications and Implementation* section, *Power Supply Recommendations* section, *Layout* section, *Device and Documentation Support* section, and *Mechanical, Packaging, and Orderable Information* section | 1 |

## 11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.


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## PACKAGING INFORMATION

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| 5962-8775401EA | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8775401EA CD54HC4053F3A |
| 5962-8855601EA | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8855601EA CD54HC4052F3A |
| 5962-9065401MEA | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9065401MEA CD54HCT4051F3A |
| CD54HC4051F | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | CD54HC4051F |
| CD54HC4051F.A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | CD54HC4051F |
| CD54HC4051F3A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | CD54HC4051F3A |
| CD54HC4051F3A.A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | CD54HC4051F3A |
| CD54HC4052F | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | CD54HC4052F |
| CD54HC4052F.A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | CD54HC4052F |
| CD54HC4052F3A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8855601EA CD54HC4052F3A |
| CD54HC4052F3A.A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8855601EA CD54HC4052F3A |
| CD54HC4053F | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | CD54HC4053F |
| CD54HC4053F.A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | CD54HC4053F |
| CD54HC4053F3A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8775401EA CD54HC4053F3A |
| CD54HC4053F3A.A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8775401EA CD54HC4053F3A |
| CD54HCT4051F3A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9065401MEA CD54HCT4051F3A |
| CD54HCT4051F3A.A | Active | Production | CDIP (J) \| 16 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9065401MEA CD54HCT4051F3A |
| CD74HC4051E | Active | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HC4051E |
| CD74HC4051E.A | Active | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HC4051E |
| CD74HC4051EE4 | Active | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HC4051E |

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| CD74HC4051M | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4051M |
| CD74HC4051M96 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4051M |
| CD74HC4051M96.A | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4051M |
| CD74HC4051M96G3 | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4051M |
| CD74HC4051M96G4 | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4051M |
| CD74HC4051MT | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4051M |
| CD74HC4051NSR | Active | Production | SOP (NS) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4051M |
| CD74HC4051NSR.A | Active | Production | SOP (NS) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4051M |
| CD74HC4051NSRE4 | Active | Production | SOP (NS) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4051M |
| CD74HC4051PWR | Active | Production | TSSOP (PW) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU \| SN | Level-1-260C-UNLIM | -55 to 125 | HJ4051 |
| CD74HC4051PWR.A | Active | Production | TSSOP (PW) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HJ4051 |
| CD74HC4051PWRG4 | Obsolete | Production | TSSOP (PW) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HJ4051 |
| CD74HC4051PWT | Obsolete | Production | TSSOP (PW) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HJ4051 |
| CD74HC4052E | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HC4052E |
| CD74HC4052E.A | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HC4052E |
| CD74HC4052M | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4052M |
| CD74HC4052M96 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4052M |
| CD74HC4052M96.A | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4052M |
| CD74HC4052M96G4 | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4052M |
| CD74HC4052MT | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4052M |
| CD74HC4052NSR | Active | Production | SOP (NS) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4052M |
| CD74HC4052NSR.A | Active | Production | SOP (NS) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4052M |
| CD74HC4052PW | Obsolete | Production | TSSOP (PW) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HJ4052 |
| CD74HC4052PWR | Active | Production | TSSOP (PW) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU \| SN | Level-1-260C-UNLIM | -55 to 125 | HJ4052 |
| CD74HC4052PWR.A | Active | Production | TSSOP (PW) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HJ4052 |
| CD74HC4052PWRG4 | Obsolete | Production | TSSOP (PW) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HJ4052 |
| CD74HC4053E | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HC4053E |
| CD74HC4053E.A | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HC4053E |
| CD74HC4053M | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4053M |
| CD74HC4053M96 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU \| SN | Level-1-260C-UNLIM | -55 to 125 | HC4053M |
| CD74HC4053M96.A | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4053M |

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| CD74HC4053M96G3 | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4053M |
| CD74HC4053M96G4 | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4053M |
| CD74HC4053MT | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HC4053M |
| CD74HC4053NSR | NRND | Production | SOP (NS) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4053M |
| CD74HC4053NSR.A | NRND | Production | SOP (NS) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HC4053M |
| CD74HC4053PWR | Active | Production | TSSOP (PW) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HJ4053 |
| CD74HC4053PWR.A | Active | Production | TSSOP (PW) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HJ4053 |
| CD74HC4053PWRG4 | Obsolete | Production | TSSOP (PW) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HJ4053 |
| CD74HC4053PWT | Obsolete | Production | TSSOP (PW) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HJ4053 |
| CD74HCT4051E | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HCT4051E |
| CD74HCT4051E.A | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HCT4051E |
| CD74HCT4051M | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HCT4051M |
| CD74HCT4051M96 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4051M |
| CD74HCT4051M96.A | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4051M |
| CD74HCT4051M96E4 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4051M |
| CD74HCT4051M96G4 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4051M |
| CD74HCT4051MT | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HCT4051M |
| CD74HCT4052E | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HCT4052E |
| CD74HCT4052E.A | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HCT4052E |
| CD74HCT4052EE4 | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HCT4052E |
| CD74HCT4052M | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HCT4052M |
| CD74HCT4052M96 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4052M |
| CD74HCT4052M96.A | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4052M |
| CD74HCT4052M96G4 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4052M |
| CD74HCT4052MT | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HCT4052M |
| CD74HCT4053E | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HCT4053E |
| CD74HCT4053E.A | NRND | Production | PDIP (N) \| 16 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -55 to 125 | CD74HCT4053E |
| CD74HCT4053M | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HCT4053M |
| CD74HCT4053M96 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4053M |
| CD74HCT4053M96.A | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4053M |
| CD74HCT4053M96E4 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4053M |

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| CD74HCT4053M96G4 | Active | Production | SOIC (D) \| 16 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HCT4053M |
| CD74HCT4053MT | Obsolete | Production | SOIC (D) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HCT4053M |
| CD74HCT4053PWR | Active | Production | TSSOP (PW) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HK4053 |
| CD74HCT4053PWR.A | Active | Production | TSSOP (PW) \| 16 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -55 to 125 | HK4053 |
| CD74HCT4053PWRG4 | Obsolete | Production | TSSOP (PW) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HK4053 |
| CD74HCT4053PWT | Obsolete | Production | TSSOP (PW) \| 16 | - | - | Call TI | Call TI | -55 to 125 | HK4053 |

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

**Important Information and Disclaimer:**The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

**OTHER QUALIFIED VERSIONS OF CD54HC4051, CD54HC4052, CD54HC4053, CD54HCT4051, CD74HC4051, CD74HC4052, CD74HC4053, CD74HCT4051 :**

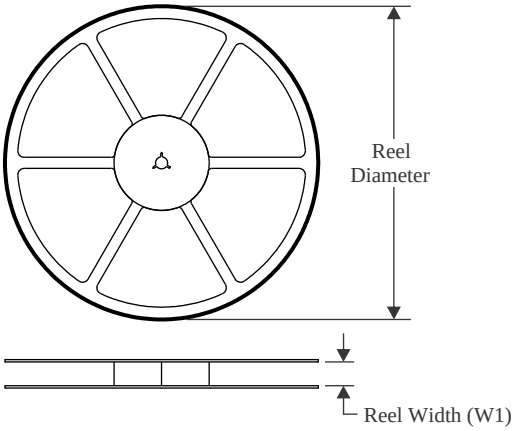
- Catalog : CD74HC4051, CD74HC4052, CD74HC4053, CD74HCT4051
- Automotive : CD74HC4051-Q1, CD74HCT4051-Q1, CD74HC4051-Q1, CD74HCT4051-Q1
- Enhanced Product : CD74HC4051-EP, CD74HC4051-EP
- Military : CD54HC4051, CD54HC4052, CD54HC4053, CD54HCT4051

NOTE: Qualified Version Definitions:

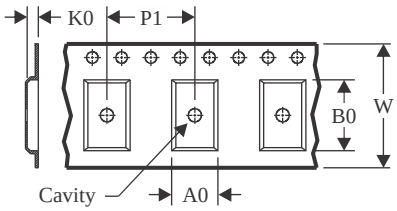
- Catalog - TI's standard catalog product
- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects
- Enhanced Product - Supports Defense, Aerospace and Medical Applications
- Military - QML certified for Military and Defense Applications

## TAPE AND REEL INFORMATION

**REEL DIMENSIONS**

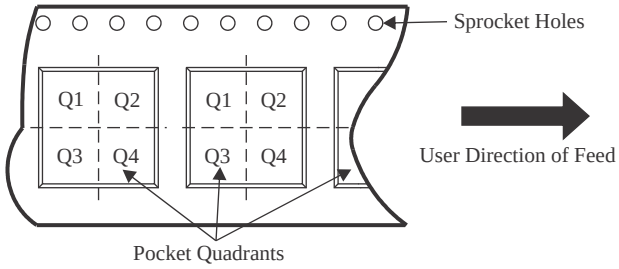



**TAPE DIMENSIONS**




| | |
|---|---|
| A0 | Dimension designed to accommodate the component width |
| B0 | Dimension designed to accommodate the component length |
| K0 | Dimension designed to accommodate the component thickness |
| W | Overall width of the carrier tape |
| P1 | Pitch between successive cavity centers |

**QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE**

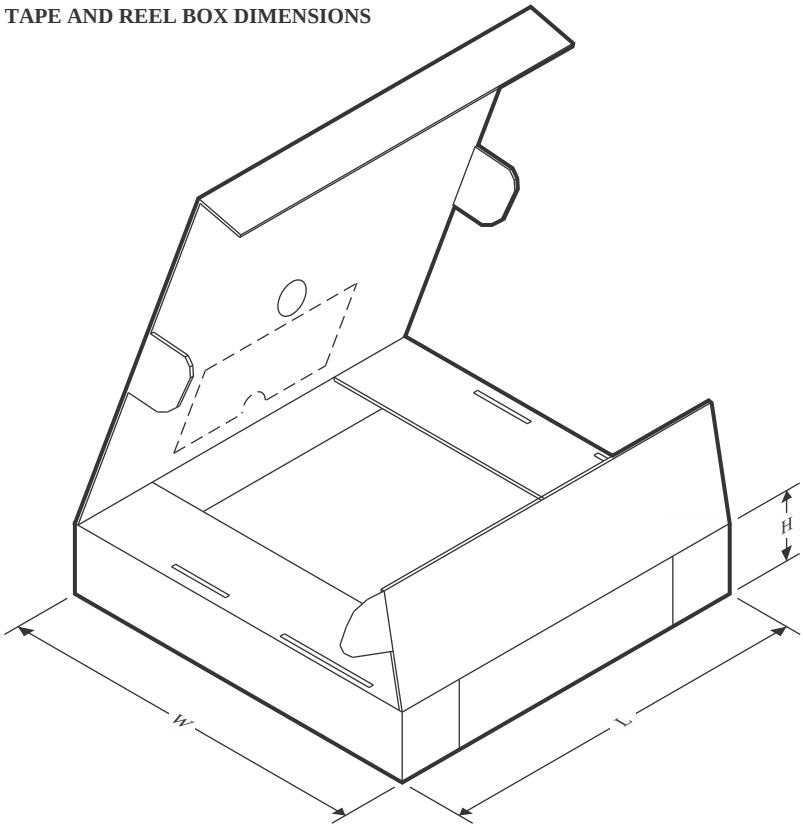



*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CD74HC4051M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HC4051M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HC4051NSR | SOP | NS | 16 | 2000 | 330.0 | 16.4 | 8.1 | 10.4 | 2.5 | 12.0 | 16.0 | Q1 |
| CD74HC4051PWR | TSSOP | PW | 16 | 2000 | 330.0 | 12.4 | 6.9 | 5.3 | 1.6 | 8.0 | 12.0 | Q1 |
| CD74HC4051PWR | TSSOP | PW | 16 | 2000 | 330.0 | 12.4 | 6.9 | 5.6 | 1.6 | 8.0 | 12.0 | Q1 |
| CD74HC4051PWR | TSSOP | PW | 16 | 2000 | 330.0 | 12.4 | 6.9 | 5.6 | 1.6 | 8.0 | 12.0 | Q1 |
| CD74HC4052M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HC4052M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HC4052NSR | SOP | NS | 16 | 2000 | 330.0 | 16.4 | 8.1 | 10.4 | 2.5 | 12.0 | 16.0 | Q1 |
| CD74HC4052PWR | TSSOP | PW | 16 | 2000 | 330.0 | 12.4 | 6.9 | 5.6 | 1.6 | 8.0 | 12.0 | Q1 |
| CD74HC4052PWR | TSSOP | PW | 16 | 2000 | 330.0 | 12.4 | 6.9 | 5.3 | 1.6 | 8.0 | 12.0 | Q1 |
| CD74HC4052PWR | TSSOP | PW | 16 | 2000 | 330.0 | 12.4 | 6.9 | 5.6 | 1.6 | 8.0 | 12.0 | Q1 |
| CD74HC4053M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HC4053M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HC4053NSR | SOP | NS | 16 | 2000 | 330.0 | 16.4 | 8.1 | 10.4 | 2.5 | 12.0 | 16.0 | Q1 |
| CD74HC4053PWR | TSSOP | PW | 16 | 2000 | 330.0 | 12.4 | 6.9 | 5.6 | 1.6 | 8.0 | 12.0 | Q1 |

| Device | Package Type | Package Drawing | Pins | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CD74HCT4051M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HCT4051M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HCT4052M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HCT4052M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HCT4053M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HCT4053M96 | SOIC | D | 16 | 2500 | 330.0 | 16.4 | 6.5 | 10.3 | 2.1 | 8.0 | 16.0 | Q1 |
| CD74HCT4053PWR | TSSOP | PW | 16 | 2000 | 330.0 | 12.4 | 6.9 | 5.6 | 1.6 | 8.0 | 12.0 | Q1 |

TAPE AND REEL BOX DIMENSIONS

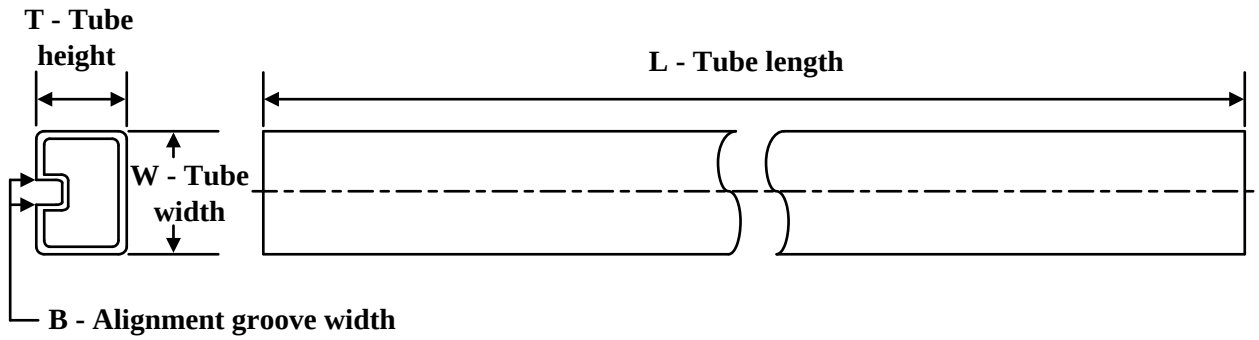



*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Length (mm) | Width (mm) | Height (mm) |
|---|---|---|---|---|---|---|---|
| CD74HC4051M96 | SOIC | D | 16 | 2500 | 340.5 | 336.1 | 32.0 |
| CD74HC4051M96 | SOIC | D | 16 | 2500 | 353.0 | 353.0 | 32.0 |
| CD74HC4051NSR | SOP | NS | 16 | 2000 | 353.0 | 353.0 | 32.0 |
| CD74HC4051PWR | TSSOP | PW | 16 | 2000 | 367.0 | 367.0 | 35.0 |
| CD74HC4051PWR | TSSOP | PW | 16 | 2000 | 356.0 | 356.0 | 35.0 |
| CD74HC4051PWR | TSSOP | PW | 16 | 2000 | 353.0 | 353.0 | 32.0 |
| CD74HC4052M96 | SOIC | D | 16 | 2500 | 353.0 | 353.0 | 32.0 |
| CD74HC4052M96 | SOIC | D | 16 | 2500 | 340.5 | 336.1 | 32.0 |
| CD74HC4052NSR | SOP | NS | 16 | 2000 | 353.0 | 353.0 | 32.0 |
| CD74HC4052PWR | TSSOP | PW | 16 | 2000 | 353.0 | 353.0 | 32.0 |
| CD74HC4052PWR | TSSOP | PW | 16 | 2000 | 367.0 | 367.0 | 35.0 |
| CD74HC4052PWR | TSSOP | PW | 16 | 2000 | 356.0 | 356.0 | 35.0 |
| CD74HC4053M96 | SOIC | D | 16 | 2500 | 353.0 | 353.0 | 32.0 |
| CD74HC4053M96 | SOIC | D | 16 | 2500 | 340.5 | 336.1 | 32.0 |
| CD74HC4053NSR | SOP | NS | 16 | 2000 | 353.0 | 353.0 | 32.0 |
| CD74HC4053PWR | TSSOP | PW | 16 | 2000 | 353.0 | 353.0 | 32.0 |
| CD74HCT4051M96 | SOIC | D | 16 | 2500 | 353.0 | 353.0 | 32.0 |
| CD74HCT4051M96 | SOIC | D | 16 | 2500 | 340.5 | 336.1 | 32.0 |

| Device | Package Type | Package Drawing | Pins | SPQ | Length (mm) | Width (mm) | Height (mm) |
|---|---|---|---|---|---|---|---|
| CD74HCT4052M96 | SOIC | D | 16 | 2500 | 353.0 | 353.0 | 32.0 |
| CD74HCT4052M96 | SOIC | D | 16 | 2500 | 340.5 | 336.1 | 32.0 |
| CD74HCT4053M96 | SOIC | D | 16 | 2500 | 353.0 | 353.0 | 32.0 |
| CD74HCT4053M96 | SOIC | D | 16 | 2500 | 340.5 | 336.1 | 32.0 |
| CD74HCT4053PWR | TSSOP | PW | 16 | 2000 | 353.0 | 353.0 | 32.0 |

## TUBE



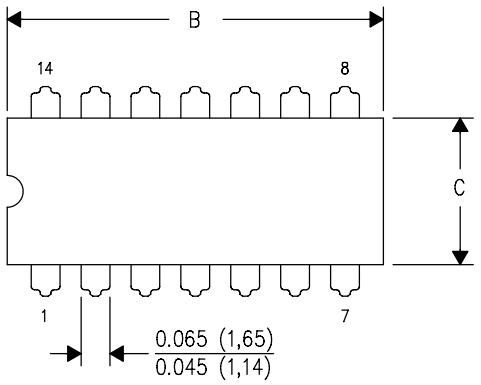

*All dimensions are nominal

| Device | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (µm) | B (mm) |
|---|---|---|---|---|---|---|---|---|
| CD74HC4051E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4051E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4051EE4 | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4052E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4052E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4052E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4052E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4053E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4053E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4053E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HC4053E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4051E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4051E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4051E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4051E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4052E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4052E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4052E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4052E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4052EE4 | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4052EE4 | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4053E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4053E | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4053E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |
| CD74HCT4053E.A | N | PDIP | 16 | 25 | 506 | 13.97 | 11230 | 4.32 |

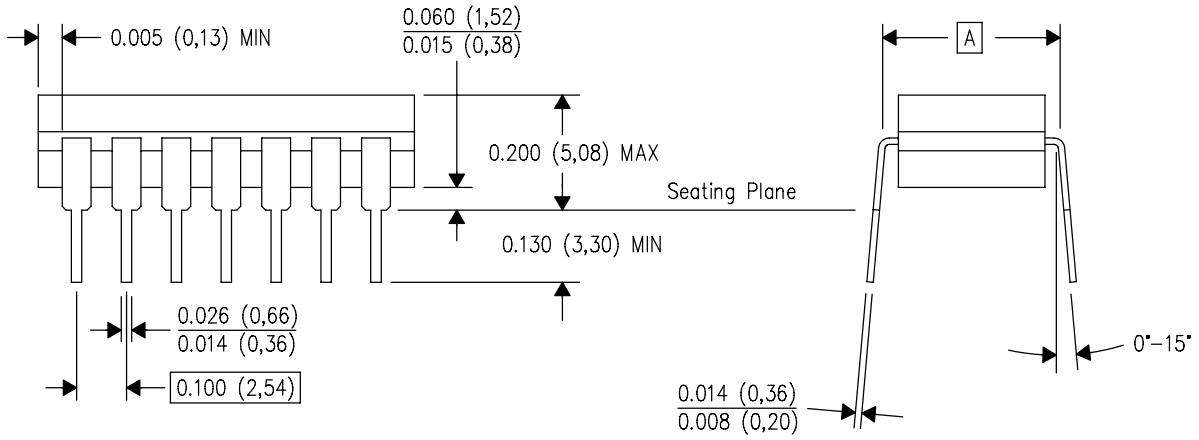
J (R–GDIP–T**)

14 LEADS SHOWN

# CERAMIC DUAL IN–LINE PACKAGE




| DIM \ PINS ** | 14 | 16 | 18 | 20 |
|---|---|---|---|---|
| A | 0.300 (7,62) BSC | 0.300 (7,62) BSC | 0.300 (7,62) BSC | 0.300 (7,62) BSC |
| B MAX | 0.785 (19,94) | .840 (21,34) | 0.960 (24,38) | 1.060 (26,92) |
| B MIN | — | — | — | — |
| C MAX | 0.300 (7,62) | 0.300 (7,62) | 0.310 (7,87) | 0.300 (7,62) |
| C MIN | 0.245 (6,22) | 0.245 (6,22) | 0.220 (5,59) | 0.245 (6,22) |




4040083/F 03/03

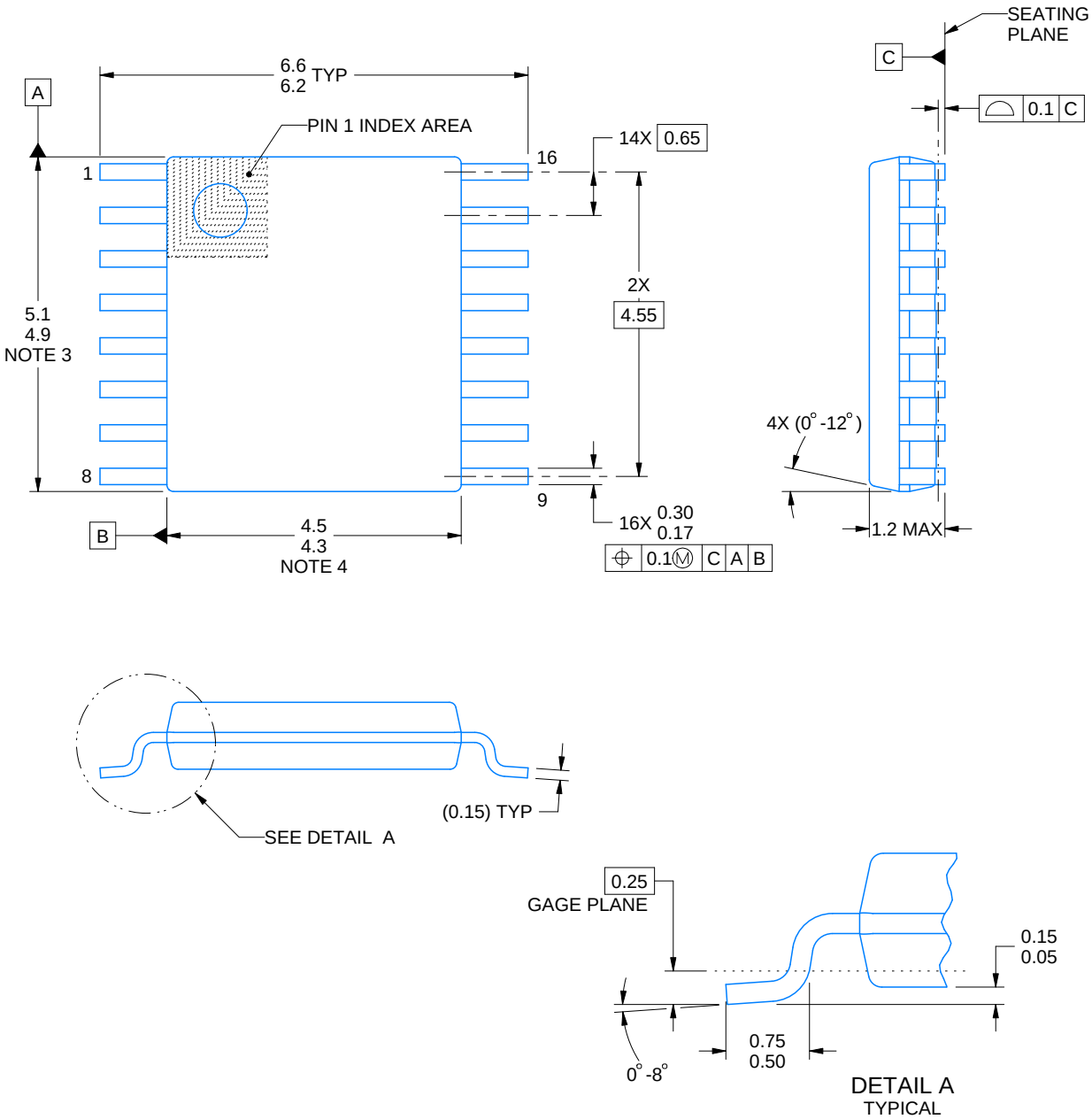
NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. This package is hermetically sealed with a ceramic lid using glass frit.
D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
E. Falls within MIL STD 1835 GDIP1–T14, GDIP1–T16, GDIP1–T18 and GDIP1–T20.

# PACKAGE OUTLINE

## PW0016A

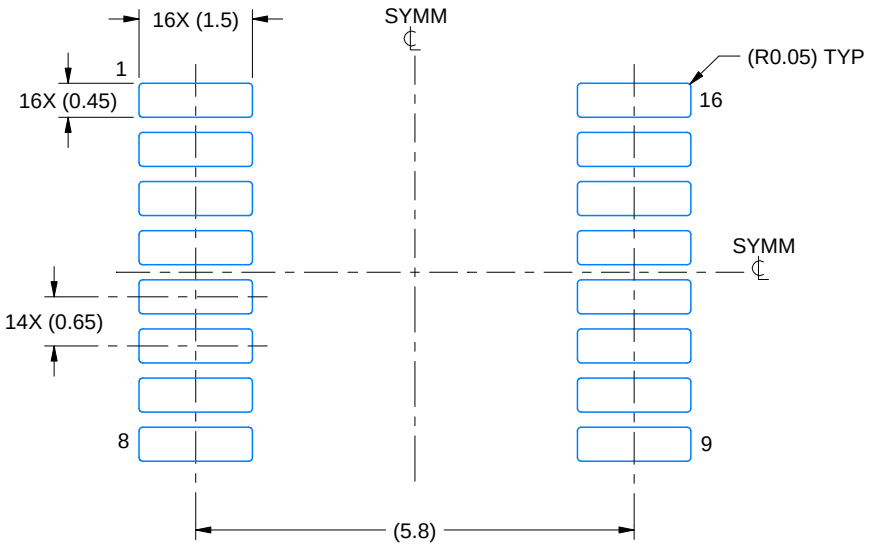
## TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE

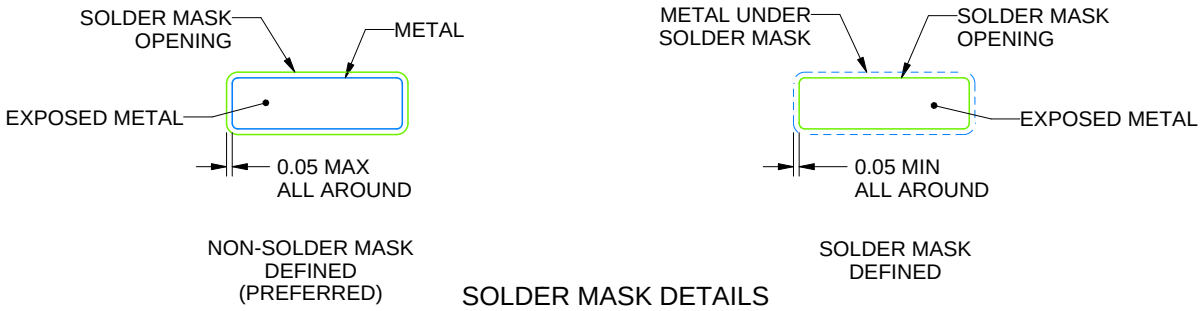



NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.




LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X

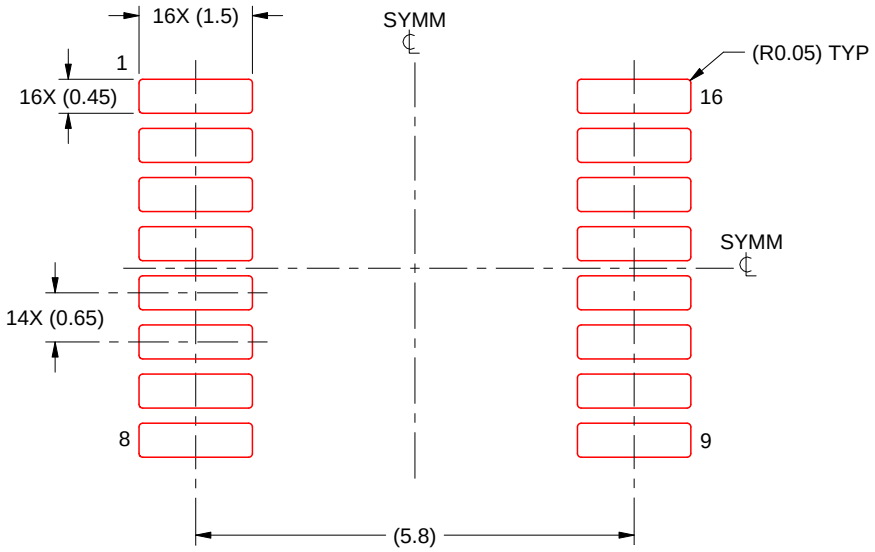



SOLDER MASK DETAILS

4220204/B 12/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220204/B 12/2023

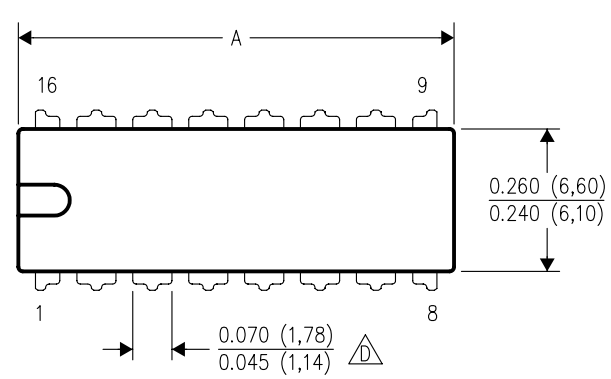
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

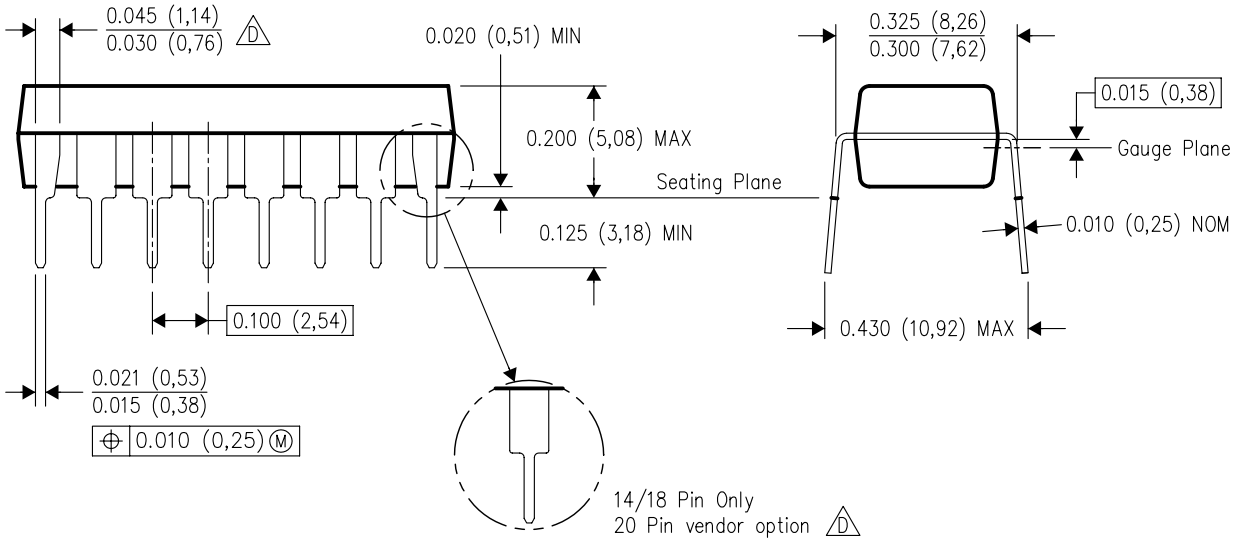
## N (R–PDIP–T**)

## PLASTIC DUAL–IN–LINE PACKAGE

16 PINS SHOWN

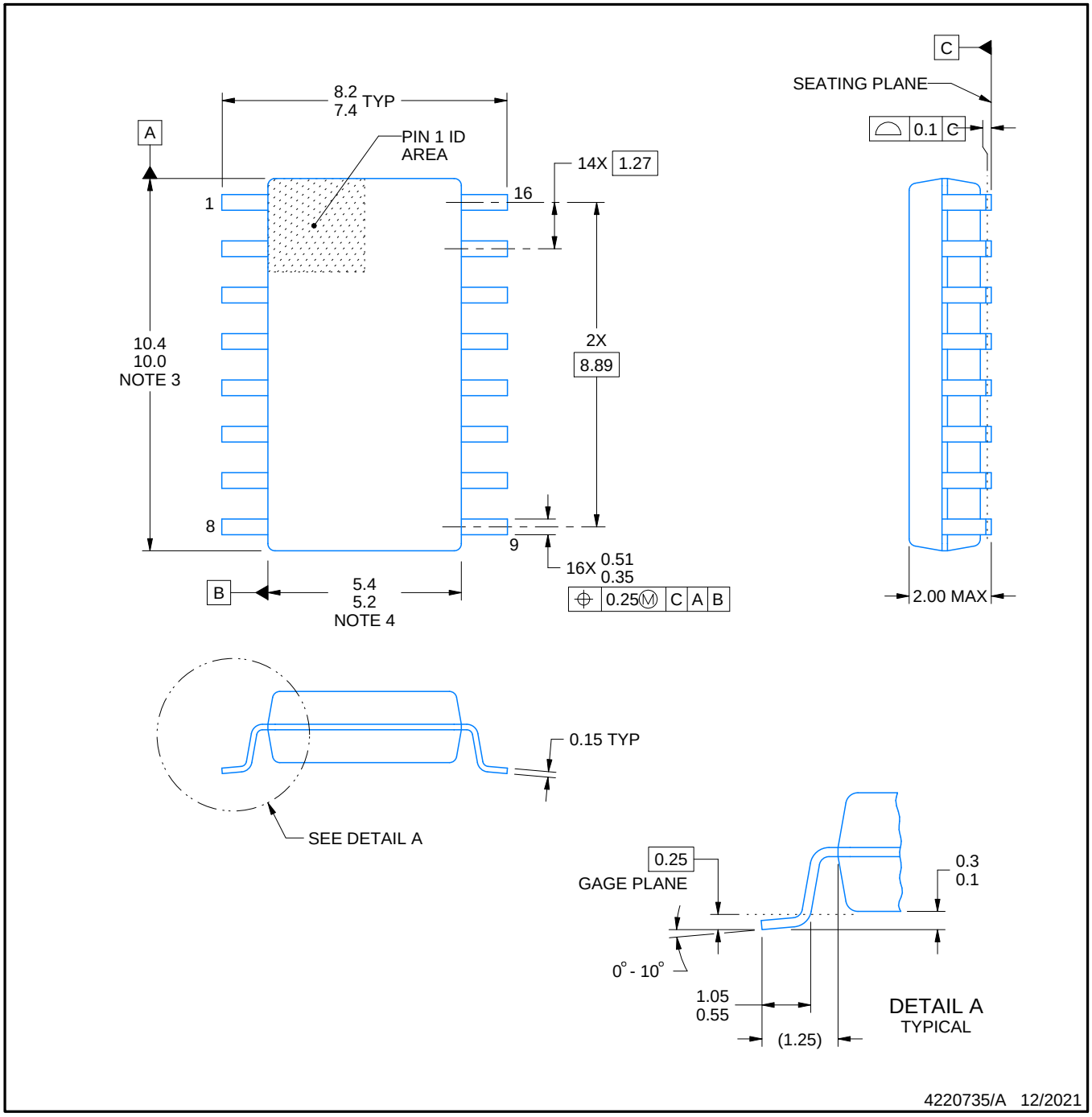



| DIM \ PINS ** | 14 | 16 | 18 | 20 |
|---|---|---|---|---|
| A MAX | 0.775 (19,69) | 0.775 (19,69) | 0.920 (23,37) | 1.060 (26,92) |
| A MIN | 0.745 (18,92) | 0.745 (18,92) | 0.850 (21,59) | 0.940 (23,88) |
| MS–001 VARIATION (C) | AA | BB | AC | AD |



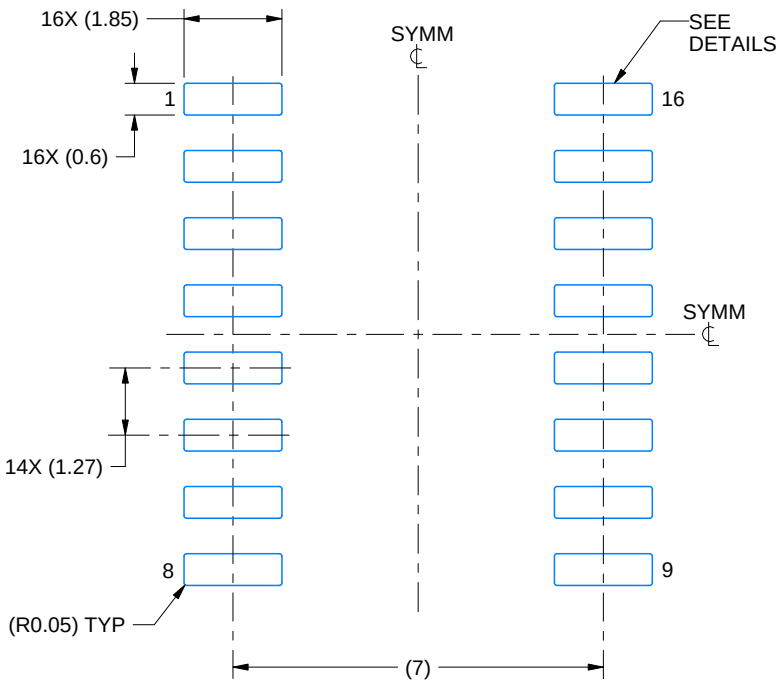

4040049/E 12/2002

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001, except 18 and 20 pin minimum body length (Dim A).
D. The 20 pin end lead shoulder width is a vendor option, either half or full width.

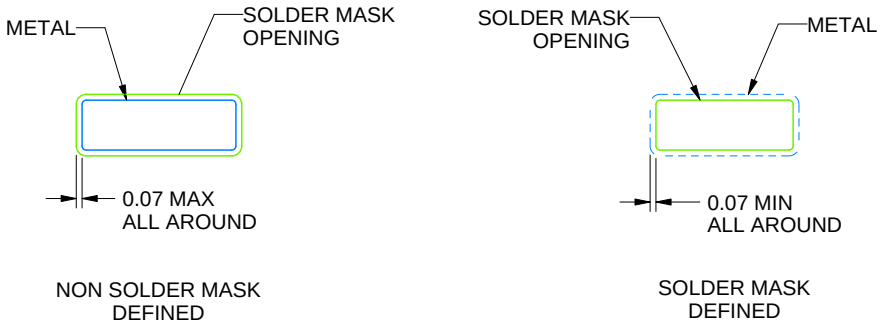



NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm, per side.




LAND PATTERN EXAMPLE
SCALE:7X

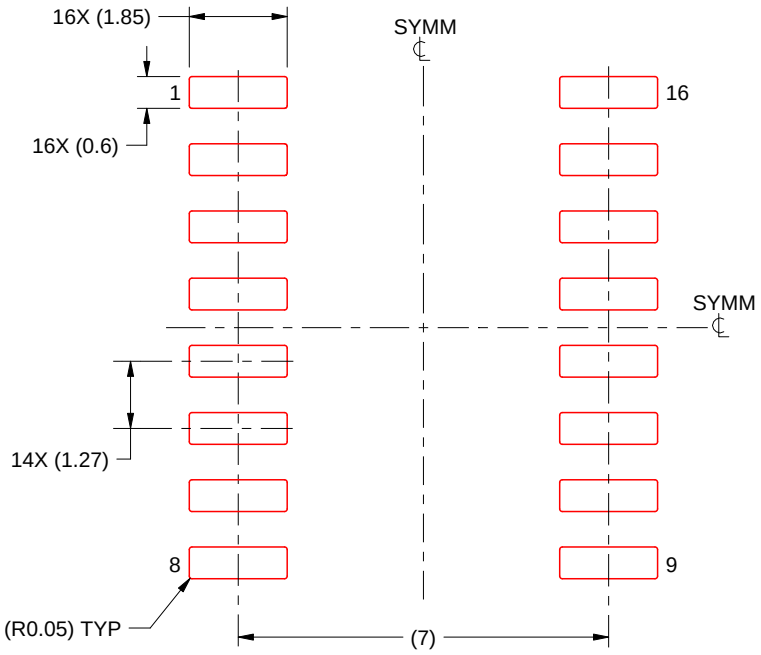



SOLDER MASK DETAILS

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NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



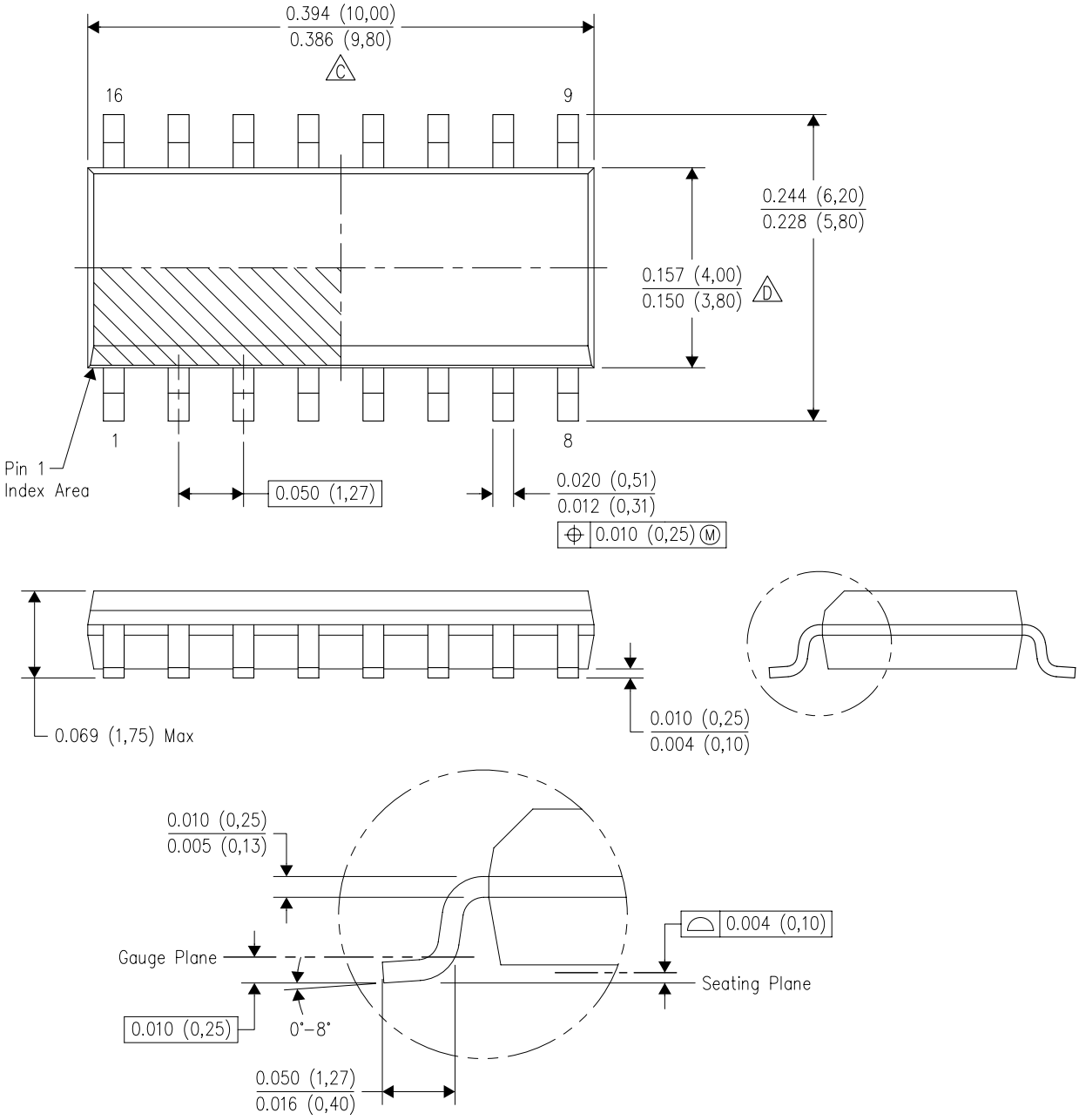

SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:7X

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NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

D (R–PDSO–G16) PLASTIC SMALL OUTLINE




NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
E. Reference JEDEC MS–012 variation AC.

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